

Introduction to Solid State Physics

EIGHTH EDITION

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*Chapter 18, Nanostructures, was written by
Professor Paul McEuen of Cornell University.*



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About the Author

Charles Kittel did his undergraduate work in physics at M.I.T and at the Cavendish Laboratory of Cambridge University. He received his Ph.D. from the University of Wisconsin. He worked in the solid state group at Bell Laboratories, along with Bardeen and Shockley, leaving to start the theoretical solid state physics group at Berkeley in 1951. His research has been largely in magnetism and in semiconductors. In magnetism he developed the theories of ferromagnetic and antiferromagnetic resonance and the theory of single ferromagnetic domains, and extended the Bloch theory of magnons. In semiconductor physics he participated in the first cyclotron and plasma resonance experiments and extended the results to the theory of impurity states and to electron-hole drops.

He has been awarded three Guggenheim fellowships, the Oliver Buckley Prize for Solid State Physics, and, for contributions to teaching, the Oersted Medal of the American Association of Physics Teachers. He is a member of the National Academy of Science and of the American Academy of Arts and Sciences.

Preface

This book is the eighth edition of an elementary text on solid state/condensed matter physics for seniors and beginning graduate students of the physical sciences, chemistry, and engineering. In the years since the first edition was published the field has developed vigorously, and there are notable applications. The challenge to the author has been to treat significant new areas while maintaining the introductory level of the text. It would be a pity to present such a physical, tactile field as an exercise in formalism.

At the first edition in 1953 superconductivity was not understood; Fermi surfaces in metals were beginning to be explored and cyclotron resonance in semiconductors had just been observed; ferrites and permanent magnets were beginning to be understood; only a few physicists then believed in the reality of spin waves. Nanophysics was forty years off. In other fields, the structure of DNA was determined and the drift of continents on the Earth was demonstrated. It was a great time to be in Science, as it is now. I have tried with the successive editions of *ISSP* to introduce new generation to the same excitement.

There are several changes from the seventh edition, as well as much clarification:

- An important chapter has been added on nanophysics, contributed by an active worker in the field, Professor Paul L. McEuen of Cornell University. Nanophysics is the science of materials with one, two, or three small dimensions, where “small” means nanometer 10^{-9} m) This field is the most exciting and vigorous addition to solid state science in the last ten years.
- The text makes use of the simplifications made possible by the universal availability of computers. Bibliographies and references have been nearly eliminated because simple computer searches using keywords on a search engine such as Google will quickly generate many useful and more recent references. As an example of what can be done on the Web, explore the entry <http://www.physicsweb.org/bestof/cond-mat>. No lack of honor is intended by the omissions of early or traditional references to the workers who first worked on the problems of the solid state.
- The order of the chapters has been changed: superconductivity and magnetism appear earlier, thereby making it easier to arrange an interesting one-semester course.

The crystallographic notation conforms with current usage in physics. Important equations in the body of the text are repeated in SI and CGS-Gaussian units, where these differ, except where a single indicated substitution will translate from CGS to SI. The dual usage in this book has been found helpful and acceptable. Tables are in conventional units. The symbol e denotes the

charge on the proton and is positive. The notation (18) refers to Equation 18 of the current chapter, but (3.18) refers to Equation 18 of Chapter 3. A caret (^) over a vector denotes a unit vector.

Few of the problems are exactly easy: Most were devised to carry forward the subject of the chapter. With few exceptions, the problems are those of the original sixth and seventh editions. The notation *QTS* refers to my *Quantum Theory of Solids, with solutions by C. Y. Fong*; *TP* refers to *Thermal Physics, with H. Kroemer*.

This edition owes much to detailed reviews of the entire text by Professor Paul L. McEuen of Cornell University and Professor Roger Lewis of Wollongong University in Australia. They helped make the book much easier to read and understand. However, I must assume responsibility for the close relation of the text to the earlier editions. Many credits for suggestions, reviews, and photographs are given in the prefaces to earlier editions. I have a great debt to Stuart Johnson, my publisher at Wiley; Suzanne Ingrao, my editor; and Barbara Bell, my personal assistant.

Corrections and suggestions will be gratefully received and may be addressed to the author by email to kittel@berkeley.edu.

The Instructor's Manual is available for download at www.wiley.com/college/kittel.

Charles Kittel

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UNITS: $1 \text{ \AA} = 1 \text{ angstrom} = 10^{-8} \text{ cm} = 0.1 \text{ nm} = 10^{-10} \text{ m}.$

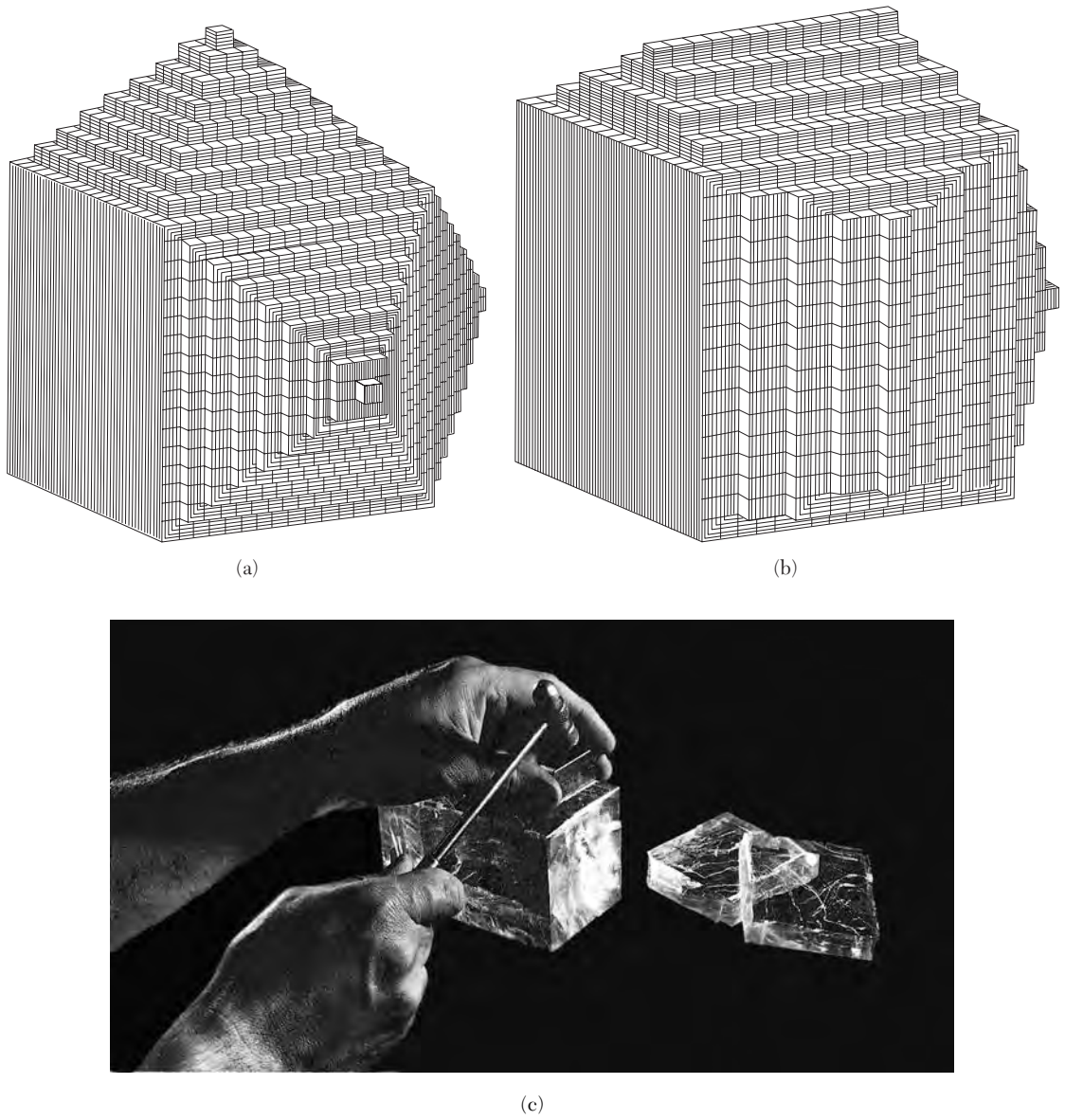


Figure 1 Relation of the external form of crystals to the form of the elementary building blocks. The building blocks are identical in (a) and (b), but different crystal faces are developed. (c) Cleaving a crystal of rock salt.

PERIODIC ARRAYS OF ATOMS

The serious study of solid state physics began with the discovery of x-ray diffraction by crystals and the publication of a series of simple calculations of the properties of crystals and of electrons in crystals. Why crystalline solids rather than noncrystalline solids? The important electronic properties of solids are best expressed in crystals. Thus the properties of the most important semiconductors depend on the crystalline structure of the host, essentially because electrons have short wavelength components that respond dramatically to the regular periodic atomic order of the specimen. Noncrystalline materials, notably glasses, are important for optical propagation because light waves have a longer wavelength than electrons and see an average over the order, and not the less regular local order itself.

We start the book with crystals. A crystal is formed by adding atoms in a constant environment, usually in a solution. Possibly the first crystal you ever saw was a natural quartz crystal grown in a slow geological process from a silicate solution in hot water under pressure. The crystal form develops as identical building blocks are added continuously. Figure 1 shows an idealized picture of the growth process, as imagined two centuries ago. The building blocks here are atoms or groups of atoms. The crystal thus formed is a three-dimensional periodic array of identical building blocks, apart from any imperfections and impurities that may accidentally be included or built into the structure.

The original experimental evidence for the periodicity of the structure rests on the discovery by mineralogists that the index numbers that define the orientations of the faces of a crystal are exact integers. This evidence was supported by the discovery in 1912 of x-ray diffraction by crystals, when Laue developed the theory of x-ray diffraction by a periodic array, and his coworkers reported the first experimental observation of x-ray diffraction by crystals. The importance of x-rays for this task is that they are waves and have a wavelength comparable with the length of a building block of the structure. Such analysis can also be done with neutron diffraction and with electron diffraction, but x-rays are usually the tool of choice.

The diffraction work proved decisively that crystals are built of a periodic array of atoms or groups of atoms. With an established atomic model of a crystal, physicists could think much further, and the development of quantum theory was of great importance to the birth of solid state physics. Related studies have been extended to noncrystalline solids and to quantum fluids. The wider field is known as condensed matter physics and is one of the largest and most vigorous areas of physics.

Lattice Translation Vectors

An ideal crystal is constructed by the infinite repetition of identical groups of atoms (Fig. 2). A group is called the **basis**. The set of mathematical points to which the basis is attached is called the **lattice**. The lattice in three dimensions may be defined by three translation vectors $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, such that the arrangement of atoms in the crystal looks the same when viewed from the point $\mathbf{r}$ as when viewed from every point $\mathbf{r}'$ translated by an integral multiple of the $\mathbf{a}$'s:

$$\mathbf{r}' = \mathbf{r} + u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + u_3\mathbf{a}_3. \quad (1)$$

Here u_1 , u_2 , u_3 are arbitrary integers. The set of points $\mathbf{r}'$ defined by (1) for all u_1 , u_2 , u_3 defines the lattice.

The lattice is said to be **primitive** if any two points from which the atomic arrangement looks the same always satisfy (1) with a suitable choice of the integers u_i . This statement defines the **primitive translation vectors** $\mathbf{a}_i$. There is no cell of smaller volume than $\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3$ that can serve as a building block for the crystal structure. We often use the primitive translation vectors to define the **crystal axes**, which form three adjacent edges of the primitive parallelepiped. Nonprimitive axes are often used as crystal axes when they have a simple relation to the symmetry of the structure.

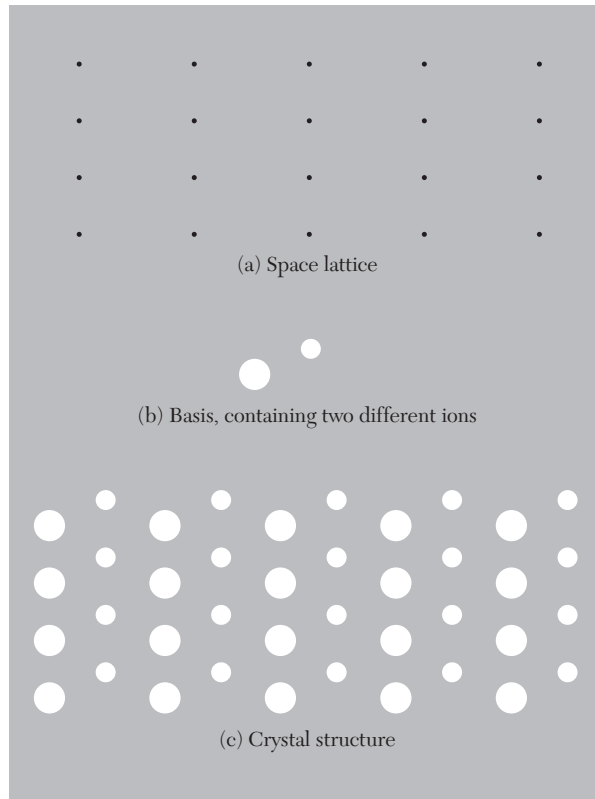


Figure 2 The crystal structure is formed by the addition of the basis (b) to every lattice point of the space lattice (a). By looking at (c), one can recognize the basis and then one can abstract the space lattice. It does not matter where the basis is put in relation to a lattice point.

Basis and the Crystal Structure

The **basis** of the crystal structure can be identified once the crystal axes have been chosen. Figure 2 shows how a crystal is made by adding a basis to every lattice point—of course the lattice points are just mathematical constructions. Every basis in a given crystal is identical to every other in composition, arrangement, and orientation.

The number of atoms in the basis may be one, or it may be more than one. The position of the center of an atom j of the basis relative to the associated lattice point is

$$\mathbf{r}_j = x_j \mathbf{a}_1 + y_j \mathbf{a}_2 + z_j \mathbf{a}_3. \quad (2)$$

We may arrange the origin, which we have called the associated lattice point, so that $0 \leq x_j, y_j, z_j \leq 1$.

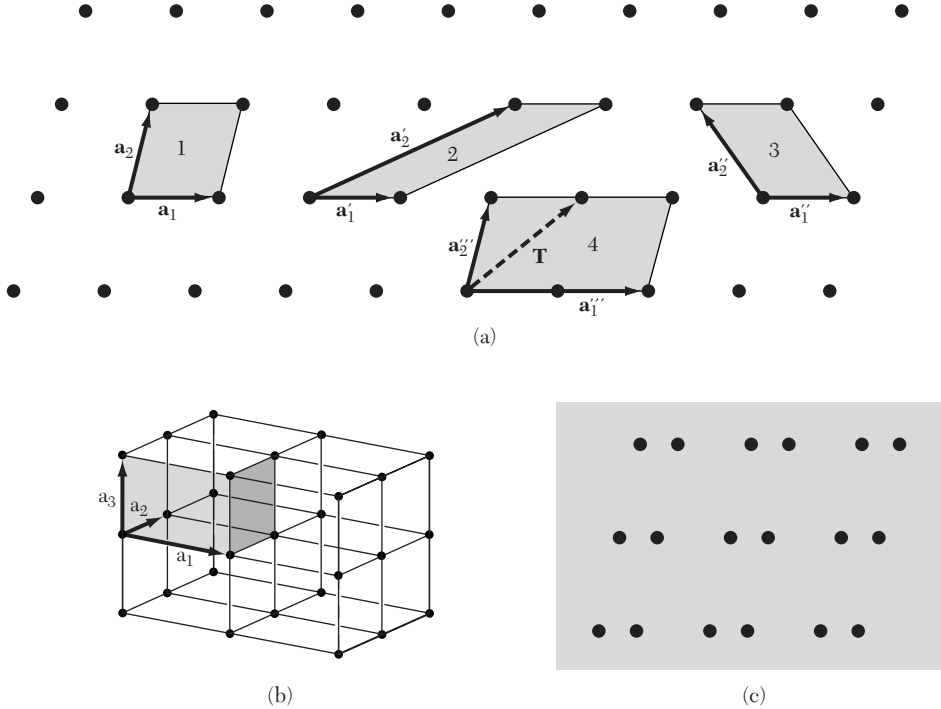
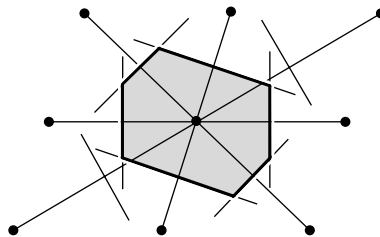


Figure 3a Lattice points of a space lattice in two dimensions. All pairs of vectors $\mathbf{a}_1, \mathbf{a}_2$ are translation vectors of the lattice. But $\mathbf{a}_1'', \mathbf{a}_2''$ are not primitive translation vectors because we cannot form the lattice translation $\mathbf{T}$ from integral combinations of $\mathbf{a}_1''$ and $\mathbf{a}_2''$. The other pairs shown of $\mathbf{a}_1$ and $\mathbf{a}_2$ may be taken as the primitive translation vectors of the lattice. The parallelograms 1, 2, 3 are equal in area and any of them could be taken as the primitive cell. The parallelogram 4 has twice the area of a primitive cell.

Figure 3b Primitive cell of a space lattice in three dimensions.

Figure 3c Suppose these points are identical atoms: Sketch in on the figure a set of lattice points, a choice of primitive axes, a primitive cell, and the basis of atoms associated with a lattice point.

Figure 4 A primitive cell may also be chosen following this procedure: (1) draw lines to connect a given lattice point to all nearby lattice points; (2) at the midpoint and normal to these lines, draw new lines or planes. The smallest volume enclosed in this way is the Wigner-Seitz primitive cell. All space may be filled by these cells, just as by the cells of Fig. 3.



Primitive Lattice Cell

The parallelepiped defined by primitive axes $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ is called a **primitive cell** (Fig. 3b). A primitive cell is a type of cell or unit cell. (The adjective unit is superfluous and not needed.) A cell will fill all space by the repetition of suitable crystal translation operations. A primitive cell is a minimum-volume cell. There are many ways of choosing the primitive axes and primitive cell for a given lattice. The number of atoms in a primitive cell or primitive basis is always the same for a given crystal structure.

There is always one lattice point per primitive cell. If the primitive cell is a parallelepiped with lattice points at each of the eight corners, each lattice point is shared among eight cells, so that the total number of lattice points in the cell is one: $8 \times \frac{1}{8} = 1$. The volume of a parallelepiped with axes $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ is

$$V_c = |\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3|, \quad (3)$$

by elementary vector analysis. The basis associated with a primitive cell is called a primitive basis. No basis contains fewer atoms than a primitive basis contains. Another way of choosing a primitive cell is shown in Fig. 4. This is known to physicists as a **Wigner-Seitz cell**.

FUNDAMENTAL TYPES OF LATTICES

Crystal lattices can be carried or mapped into themselves by the lattice translations $\mathbf{T}$ and by various other symmetry operations. A typical symmetry operation is that of rotation about an axis that passes through a lattice point. Lattices can be found such that one-, two-, three-, four-, and sixfold rotation axes carry the lattice into itself, corresponding to rotations by 2π , $2\pi/2$, $2\pi/3$, $2\pi/4$, and $2\pi/6$ radians and by integral multiples of these rotations. The rotation axes are denoted by the symbols 1, 2, 3, 4, and 6.

We cannot find a lattice that goes into itself under other rotations, such as by $2\pi/7$ radians or $2\pi/5$ radians. A single molecule properly designed can have any degree of rotational symmetry, but an infinite periodic lattice cannot. We can make a crystal from molecules that individually have a fivefold rotation axis, but we should not expect the lattice to have a fivefold rotation axis. In Fig. 5 we show what happens if we try to construct a periodic lattice having fivefold

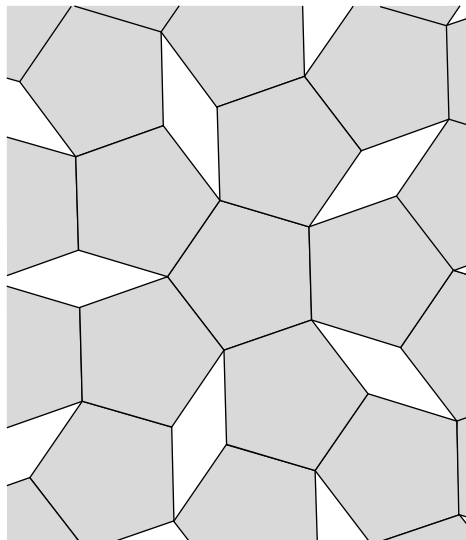
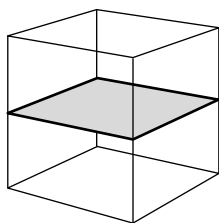
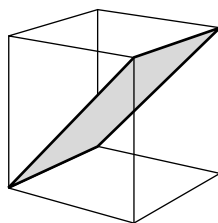


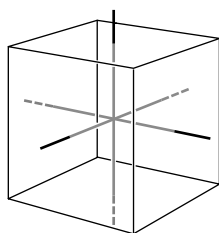
Figure 5 A fivefold axis of symmetry cannot exist in a periodic lattice because it is not possible to fill the area of a plane with a connected array of pentagons. We can, however, fill all the area of a plane with just two distinct designs of “tiles” or elementary polygons.



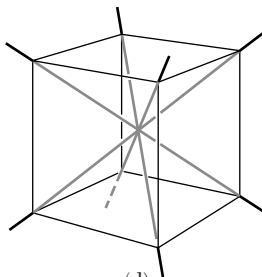
(a)



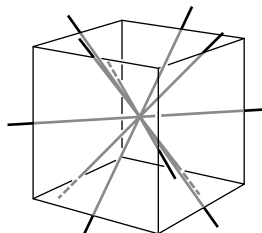
(b)



(c)



(d)



(e)

Figure 6 (a) A plane of symmetry parallel to the faces of a cube. (b) A diagonal plane of symmetry in a cube. (c) The three tetrad axes of a cube. (d) The four triad axes of a cube. (e) The six diad axes of a cube.

symmetry: the pentagons do not fit together to fill all space, showing that we cannot combine fivefold point symmetry with the required translational periodicity.

By lattice point group we mean the collection of symmetry operations which, applied about a lattice point, carry the lattice into itself. The possible rotations have been listed. We can have mirror reflections m about a plane through

a lattice point. The inversion operation is composed of a rotation of π followed by reflection in a plane normal to the rotation axis; the total effect is to replace $\mathbf{r}$ by $-\mathbf{r}$. The symmetry axes and symmetry planes of a cube are shown in Fig. 6.

Two-Dimensional Lattice Types

The lattice in Fig. 3a was drawn for arbitrary $\mathbf{a}_1$ and $\mathbf{a}_2$. A general lattice such as this is known as an **oblique lattice** and is invariant only under rotation of π and 2π about any lattice point. But special lattices of the oblique type can be invariant under rotation of $2\pi/3$, $2\pi/4$, or $2\pi/6$, or under mirror reflection. We must impose restrictive conditions on $\mathbf{a}_1$ and $\mathbf{a}_2$ if we want to construct a lattice that will be invariant under one or more of these new operations. There are four distinct types of restriction, and each leads to what we may call a **special lattice type**. Thus there are five distinct lattice types in two dimensions, the oblique lattice and the four special lattices shown in Fig. 7. **Bravais lattice** is the common phrase for a distinct lattice type; we say that there are five Bravais lattices in two dimensions.

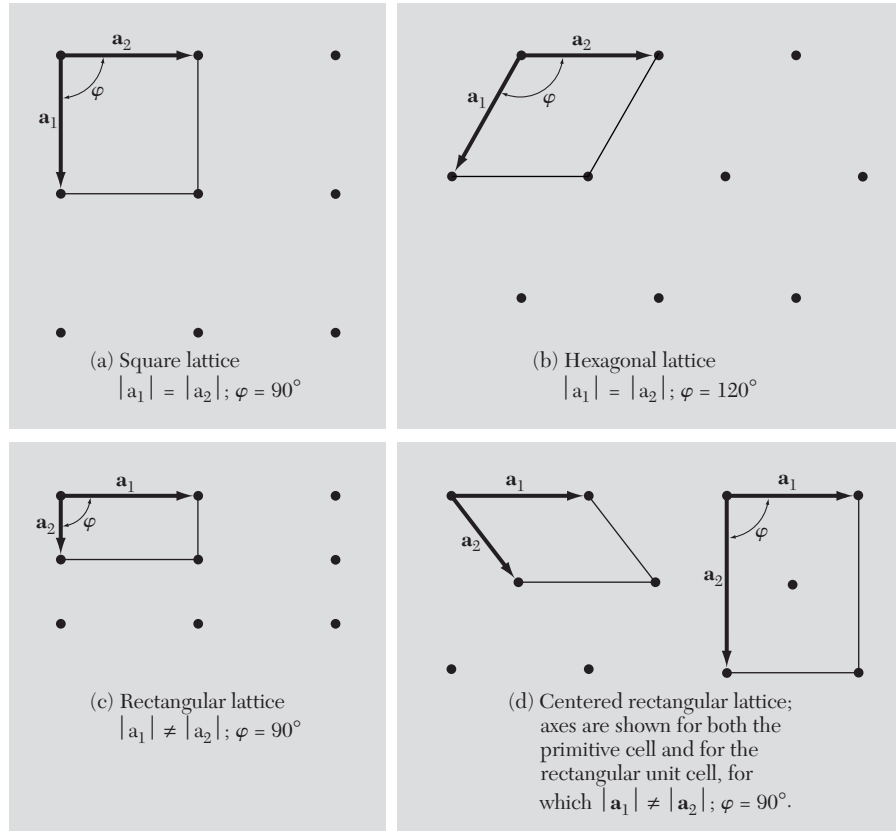


Figure 7 Four special lattices in two dimensions.

Three-Dimensional Lattice Types

The point symmetry groups in three dimensions require the 14 different lattice types listed in Table 1. The general lattice is triclinic, and there are 13 special lattices. These are grouped for convenience into systems classified according to seven types of cells, which are triclinic, monoclinic, orthorhombic, tetragonal, cubic, trigonal, and hexagonal. The division into systems is expressed in the table in terms of the axial relations that describe the cells. The cells in Fig. 8 are conventional cells: of these only the sc is a primitive cell. Often a nonprimitive cell has a more obvious relation with the point symmetry operations than has a primitive cell.

There are three lattices in the cubic system: the simple cubic (sc) lattice, the body-centered cubic (bcc) lattice, and the face-centered cubic (fcc) lattice.

Table 1 The 14 lattice types in three dimensions

System	Number of lattices	Restrictions on conventional cell axes and angles
Triclinic	1	$a_1 \neq a_2 \neq a_3$ $\alpha \neq \beta \neq \gamma$
Monoclinic	2	$a_1 \neq a_2 \neq a_3$ $\alpha = \gamma = 90^\circ \neq \beta$
Orthorhombic	4	$a_1 \neq a_2 \neq a_3$ $\alpha = \beta = \gamma = 90^\circ$
Tetragonal	2	$a_1 = a_2 \neq a_3$ $\alpha = \beta = \gamma = 90^\circ$
Cubic	3	$a_1 = a_2 = a_3$ $\alpha = \beta = \gamma = 90^\circ$
Trigonal	1	$a_1 = a_2 = a_3$ $\alpha = \beta = \gamma < 120^\circ, \neq 90^\circ$
Hexagonal	1	$a_1 = a_2 \neq a_3$ $\alpha = \beta = 90^\circ$ $\gamma = 120^\circ$

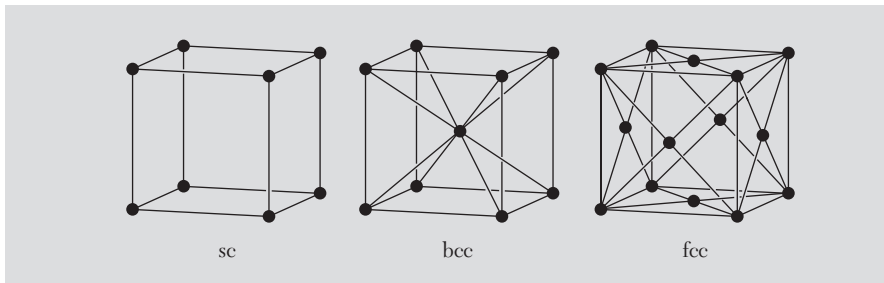


Figure 8 The cubic space lattices. The cells shown are the conventional cells.

Table 2 Characteristics of cubic lattices^a

	Simple	Body-centered	Face-centered
Volume, conventional cell	a^3	a^3	a^3
Lattice points per cell	1	2	4
Volume, primitive cell	a^3	$\frac{1}{2}a^3$	$\frac{1}{4}a^3$
Lattice points per unit volume	$1/a^3$	$2/a^3$	$4/a^3$
Number of nearest neighbors	6	8	12
Nearest-neighbor distance	a	$3^{1/2} a/2 = 0.866a$	$a/2^{1/2} = 0.707a$
Number of second neighbors	12	6	6
Second neighbor distance	$2^{1/2}a$	a	a
Packing fraction ^a	$\frac{1}{6}\pi$ =0.524	$\frac{1}{8}\pi\sqrt{3}$ =0.680	$\frac{1}{6}\pi\sqrt{2}$ =0.740

^aThe packing fraction is the maximum proportion of the available volume that can be filled with hard spheres.

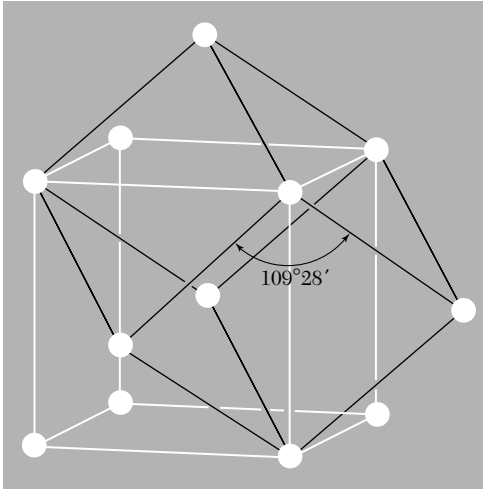


Figure 9 Body-centered cubic lattice, showing a primitive cell. The primitive cell shown is a rhombohedron of edge $\frac{1}{2}\sqrt{3}a$, and the angle between adjacent edges is $109^\circ28'$.

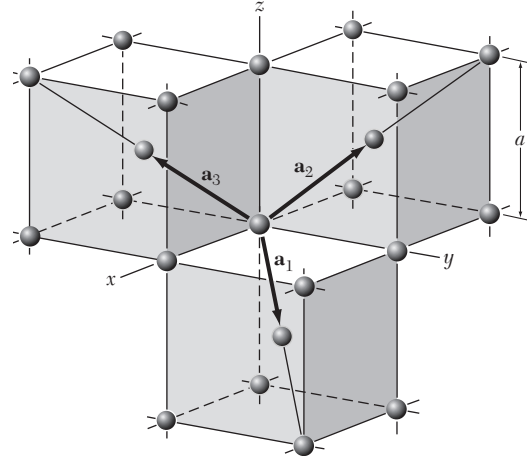


Figure 10 Primitive translation vectors of the body-centered cubic lattice; these vectors connect the lattice point at the origin to lattice points at the body centers. The primitive cell is obtained on completing the rhombohedron. In terms of the cube edge a , the primitive translation vectors are

$$\mathbf{a}_1 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}) ; \quad \mathbf{a}_2 = \frac{1}{2}a(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) ;$$

$$\mathbf{a}_3 = \frac{1}{2}a(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}) .$$

Here $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ are the Cartesian unit vectors.

The characteristics of the three cubic lattices are summarized in Table 2. A primitive cell of the bcc lattice is shown in Fig. 9, and the primitive translation vectors are shown in Fig. 10. The primitive translation vectors of the fcc lattice are shown in Fig. 11. Primitive cells by definition contain only one lattice point, but the conventional bcc cell contains two lattice points, and the fcc cell contains four lattice points.

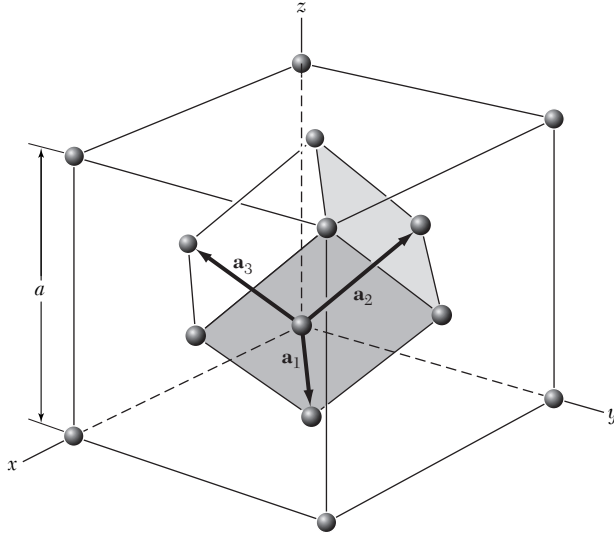


Figure 11 The rhombohedral primitive cell of the face-centered cubic crystal. The primitive translation vectors $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ connect the lattice point at the origin with lattice points at the face centers. As drawn, the primitive vectors are:

$$\mathbf{a}_1 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{y}}) ; \quad \mathbf{a}_2 = \frac{1}{2}a(\hat{\mathbf{y}} + \hat{\mathbf{z}}) ; \quad \mathbf{a}_3 = \frac{1}{2}a(\hat{\mathbf{z}} + \hat{\mathbf{x}}) .$$

The angles between the axes are 60° .

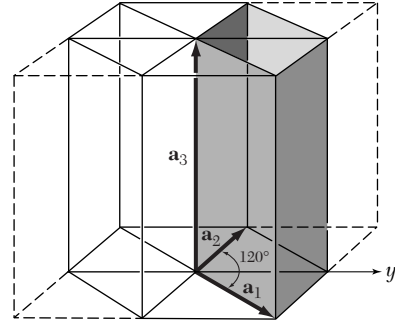


Figure 12 Relation of the primitive cell in the hexagonal system (heavy lines) to a prism of hexagonal symmetry. Here $a_1 = a_2 \neq a_3$.

The position of a point in a cell is specified by (2) in terms of the atomic coordinates x , y , z . Here each coordinate is a fraction of the axial length a_1 , a_2 , a_3 in the direction of the coordinate axis, with the origin taken at one corner of the cell. Thus the coordinates of the body center of a cell are $\frac{1}{2}\frac{1}{2}\frac{1}{2}$, and the face centers include $\frac{1}{2}\frac{1}{2}0$, $0\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}0\frac{1}{2}$. In the hexagonal system the primitive cell is a right prism based on a rhombus with an included angle of 120° . Figure 12 shows the relationship of the rhombic cell to a hexagonal prism.

INDEX SYSTEM FOR CRYSTAL PLANES

The orientation of a crystal plane is determined by three points in the plane, provided they are not collinear. If each point lay on a different crystal axis, the plane could be specified by giving the coordinates of the points in terms of the lattice constants a_1 , a_2 , a_3 . However, it turns out to be more useful for structure analysis to specify the orientation of a plane by the indices determined by the following rules (Fig. 13).

- Find the intercepts on the axes in terms of the lattice constants a_1 , a_2 , a_3 . The axes may be those of a primitive or nonprimitive cell.

Figure 13 This plane intercepts the $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ axes at $3\mathbf{a}_1$, $2\mathbf{a}_2$, $2\mathbf{a}_3$. The reciprocals of these numbers are $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{2}$. The smallest three integers having the same ratio are 2, 3, 3, and thus the indices of the plane are (233).

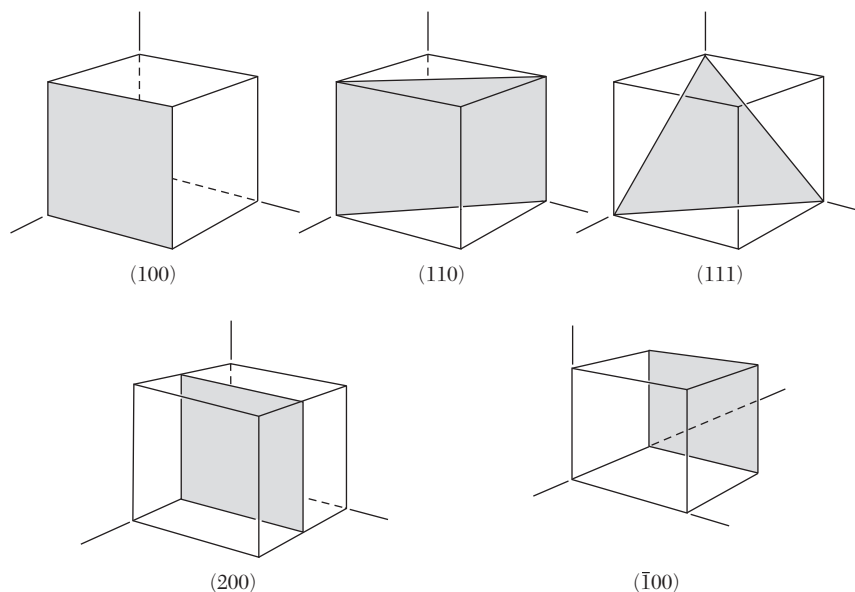
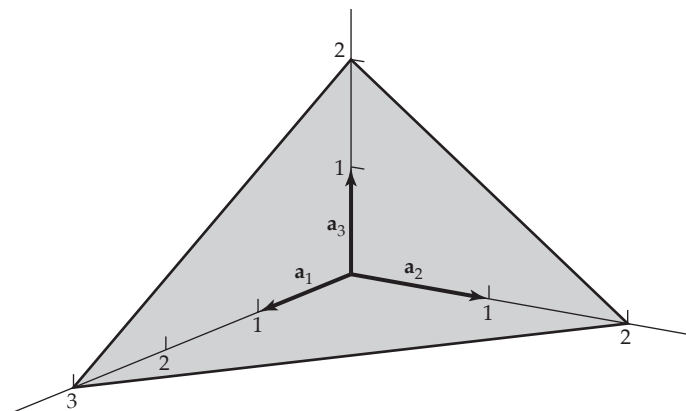


Figure 14 Indices of important planes in a cubic crystal. The plane (200) is parallel to (100) and to (100).

- Take the reciprocals of these numbers and then reduce to three integers having the same ratio, usually the smallest three integers. The result, enclosed in parentheses (hkl), is called the index of the plane.

For the plane whose intercepts are 4, 1, 2, the reciprocals are $\frac{1}{4}$, 1, and $\frac{1}{2}$; the smallest three integers having the same ratio are (142). For an intercept at infinity, the corresponding index is zero. The indices of some important planes in a cubic crystal are illustrated by Fig. 14. The indices (hkl) may denote a single plane or a set of parallel planes. If a plane cuts an axis on the negative side of the origin, the corresponding index is negative, indicated by placing a minus sign

above the index: $(h\bar{k}l)$. The cube faces of a cubic crystal are (100) , (010) , (001) , $(\bar{1}00)$, $(0\bar{1}0)$, and $(00\bar{1})$. Planes equivalent by symmetry may be denoted by curly brackets (braces) around indices; the set of cube faces is $\{100\}$. When we speak of the (200) plane we mean a plane parallel to (100) but cutting the $\mathbf{a}_1$ axis at $\frac{1}{2}a$.

The indices $[uvw]$ of a direction in a crystal are the set of the smallest integers that have the ratio of the components of a vector in the desired direction, referred to the axes. The $\mathbf{a}_1$ axis is the $[100]$ direction; the $-\mathbf{a}_2$ axis is the $[0\bar{1}0]$ direction. In cubic crystals the direction $[hkl]$ is perpendicular to a plane (hkl) having the same indices, but this is not generally true in other crystal systems.

SIMPLE CRYSTAL STRUCTURES

We discuss simple crystal structures of general interest: the sodium chloride, cesium chloride, hexagonal close-packed, diamond, and cubic zinc sulfide structures.

Sodium Chloride Structure

The sodium chloride, NaCl, structure is shown in Figs. 15 and 16. The lattice is face-centered cubic; the basis consists of one Na^+ ion and one Cl^- ion

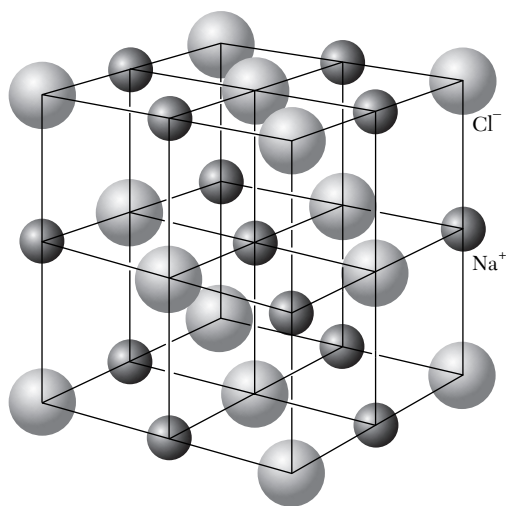


Figure 15 We may construct the sodium chloride crystal structure by arranging Na^+ and Cl^- ions alternately at the lattice points of a simple cubic lattice. In the crystal each ion is surrounded by six nearest neighbors of the opposite charge. The space lattice is fcc, and the basis has one Cl^- ion at 000 and one Na^+ ion at $\frac{1}{2}\frac{1}{2}\frac{1}{2}$. The figure shows one conventional cubic cell. The ionic diameters here are reduced in relation to the cell in order to clarify the spatial arrangement.

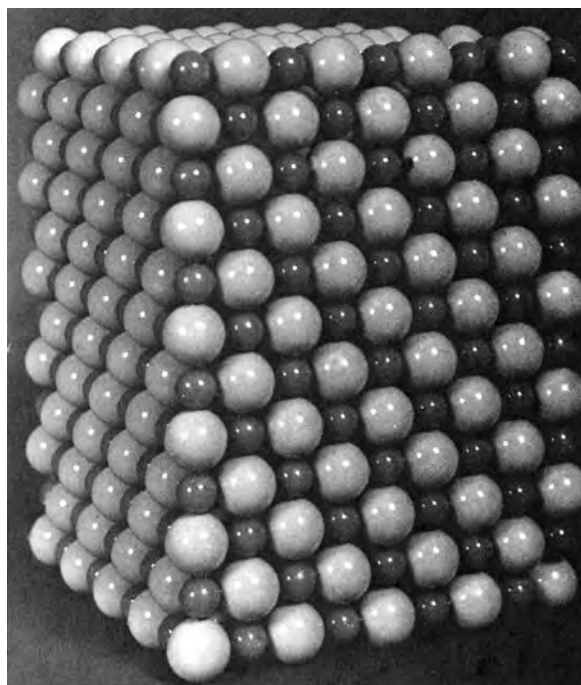


Figure 16 Model of sodium chloride. The sodium ions are smaller than the chlorine ions. (Courtesy of A. N. Holden and P. Singer.)



Figure 17 Natural crystals of lead sulfide, PbS, which has the NaCl crystal structure. (Photograph by B. Burleson.)

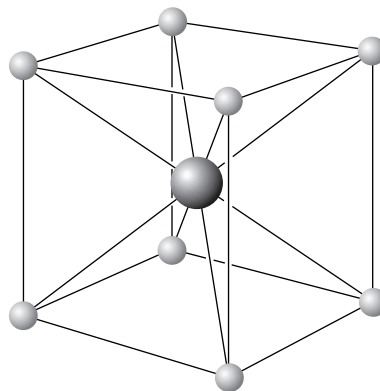


Figure 18 The cesium chloride crystal structure. The space lattice is simple cubic, and the basis has one Cs^+ ion at 000 and one Cl^- ion at $\frac{1}{2} \frac{1}{2} \frac{1}{2}$.

separated by one-half the body diagonal of a unit cube. There are four units of NaCl in each unit cube, with atoms in the positions

Cl:	000 ;	$\frac{1}{2} \frac{1}{2} 0$;	$\frac{1}{2} 0 \frac{1}{2}$;	$0 \frac{1}{2} \frac{1}{2}$.
Na:	$\frac{1}{2} \frac{1}{2} \frac{1}{2}$;	$00 \frac{1}{2}$;	$0 \frac{1}{2} 0$;	$\frac{1}{2} 00$.

Each atom has as nearest neighbors six atoms of the opposite kind. Representative crystals having the NaCl arrangement include those in the following table. The cube edge a is given in angstroms; $1 \text{ \AA} \equiv 10^{-8} \text{ cm} \equiv 10^{-10} \text{ m} \equiv 0.1 \text{ nm}$. Figure 17 is a photograph of crystals of lead sulfide (PbS) from Joplin, Missouri. The Joplin specimens form in beautiful cubes.

Crystal	a	Crystal	a
LiH	4.08 $\text{\AA}$	AgBr	5.77 $\text{\AA}$
MgO	4.20	PbS	5.92
MnO	4.43	KCl	6.29
NaCl	5.63	KBr	6.59

Cesium Chloride Structure

The cesium chloride structure is shown in Fig. 18. There is one molecule per primitive cell, with atoms at the corners 000 and body-centered positions $\frac{1}{2} \frac{1}{2} \frac{1}{2}$ of the simple cubic space lattice. Each atom may be viewed as at the center

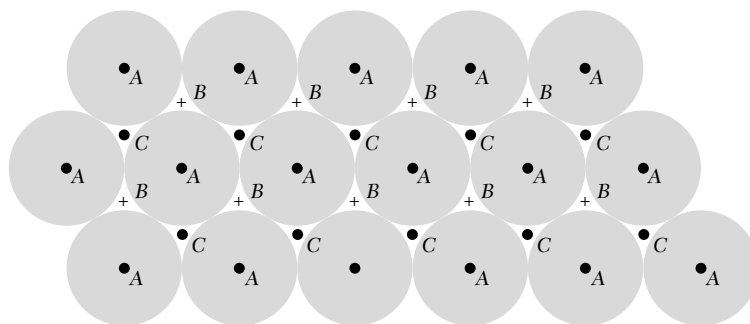


Figure 19 A close-packed layer of spheres is shown, with centers at points marked A. A second and identical layer of spheres can be placed on top of this, above and parallel to the plane of the drawing, with centers over the points marked B. There are two choices for a third layer. It can go in over A or over C. If it goes in over A, the sequence is $ABABAB \dots$ and the structure is hexagonal close-packed. If the third layer goes in over C, the sequence is $ABCABCABC \dots$ and the structure is face-centered cubic.

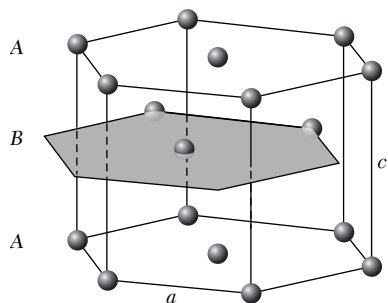


Figure 20 The hexagonal close-packed structure. The atom positions in this structure do not constitute a space lattice. The space lattice is simple hexagonal with a basis of two identical atoms associated with each lattice point. The lattice parameters a and c are indicated, where a is in the basal plane and c is the magnitude of the axis $\mathbf{a}_3$ of Fig. 12.

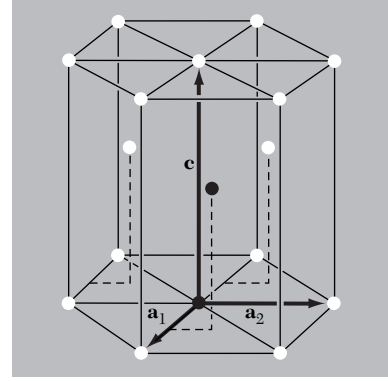
of a cube of atoms of the opposite kind, so that the number of nearest neighbors or coordination number is eight.

Crystal	a	Crystal	a
BeCu	2.70 Å	LiHg	3.29 Å
AlNi	2.88	NH ₄ Cl	3.87
CuZn (β -brass)	2.94	TlBr	3.97
CuPd	2.99	CsCl	4.11
AgMg	3.28	TlI	4.20

Hexagonal Close-Packed Structure (hcp)

There are an infinite number of ways of arranging identical spheres in a regular array that maximizes the packing fraction (Fig. 19). One is the face-centered cubic structure; another is the hexagonal close-packed structure (Fig. 20). The fraction of the total volume occupied by the spheres is 0.74 for both structures. No structure, regular or not, has denser packing.

Figure 21 The primitive cell has $a_1 = a_2$, with an included angle of 120° . The c axis (or $\mathbf{a}_3$) is normal to the plane of $\mathbf{a}_1$ and $\mathbf{a}_2$. The ideal hcp structure has $c = 1.633 a$. The two atoms of one basis are shown as solid circles. One atom of the basis is at the origin; the other atom is at $\frac{2}{3}\mathbf{a}_1 + \frac{1}{3}\mathbf{a}_2$, which means at the position $\mathbf{r} = \frac{2}{3}\mathbf{a}_1 + \frac{1}{3}\mathbf{a}_2 + \frac{1}{2}\mathbf{a}_3$.



Spheres are arranged in a single closest-packed layer *A* by placing each sphere in contact with six others in a plane. This layer may serve as either the basal plane of an hcp structure or the (111) plane of the fcc structure. A second similar layer *B* may be added by placing each sphere of *B* in contact with three spheres of the bottom layer, as in Figs. 19–21. A third layer *C* may be added in two ways. We obtain the fcc structure if the spheres of the third layer are added over the holes in the first layer that are not occupied by *B*. We obtain the hcp structure when the spheres in the third layer are placed directly over the centers of the spheres in the first layer.

The number of nearest-neighbor atoms is 12 for both hcp and fcc structures. If the binding energy (or free energy) depended only on the number of nearest-neighbor bonds per atom, there would be no difference in energy between the fcc and hcp structures.

Crystal	c/a	Crystal	c/a	Crystal	c/a
He	1.633	Zn	1.861	Zr	1.594
Be	1.581	Cd	1.886	Gd	1.592
Mg	1.623	Co	1.622	Lu	1.586
Ti	1.586	Y	1.570		

Diamond Structure

The diamond structure is the structure of the semiconductors silicon and germanium and is related to the structure of several important semiconductor binary compounds. The space lattice of diamond is face-centered cubic. The primitive basis of the diamond structure has two identical atoms at coordinates 000 and $\frac{1}{4}\frac{1}{4}\frac{1}{4}$ associated with each point of the fcc lattice, as shown in Fig. 22. Because the conventional unit cube of the fcc lattice contains 4 lattice points, it follows that the conventional unit cube of the diamond structure contains $2 \times 4 = 8$ atoms. There is no way to choose a primitive cell such that the basis of diamond contains only one atom.

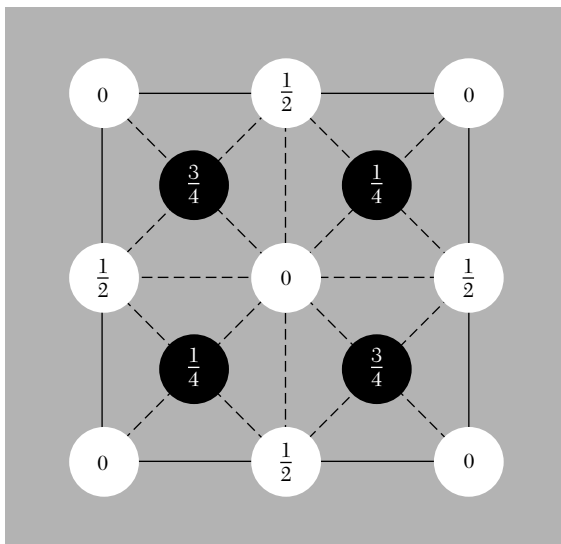


Figure 22 Atomic positions in the cubic cell of the diamond structure projected on a cube face; fractions denote height above the base in units of a cube edge. The points at 0 and $\frac{1}{2}$ are on the fcc lattice; those at $\frac{1}{4}$ and $\frac{3}{4}$ are on a similar lattice displaced along the body diagonal by one-fourth of its length. With a fcc space lattice, the basis consists of two identical atoms at 000 and $\frac{1}{4}\frac{1}{4}\frac{1}{4}$.

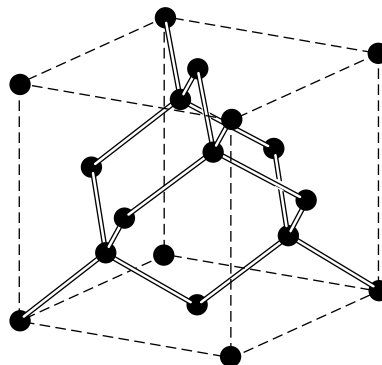


Figure 23 Crystal structure of diamond, showing the tetrahedral bond arrangement.

The tetrahedral bonding characteristic of the diamond structure is shown in Fig. 23. Each atom has 4 nearest neighbors and 12 next nearest neighbors. The diamond structure is relatively empty: the maximum proportion of the available volume which may be filled by hard spheres is only 0.34, which is 46 percent of the filling factor for a closest-packed structure such as fcc or hcp. The diamond structure is an example of the directional covalent bonding found in column IV of the periodic table of elements. Carbon, silicon, germanium, and tin can crystallize in the diamond structure, with lattice constants $a = 3.567, 5.430, 5.658, \text{ and } 6.49 \text{ \AA}$, respectively. Here a is the edge of the conventional cubic cell.

Cubic Zinc Sulfide Structure

The diamond structure may be viewed as two fcc structures displaced from each other by one-quarter of a body diagonal. The cubic zinc sulfide (zinc blende) structure results when Zn atoms are placed on one fcc lattice and S atoms on the other fcc lattice, as in Fig. 24. The conventional cell is a cube. The coordinates of the Zn atoms are 000; $0\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}0\frac{1}{2}$; $\frac{1}{2}\frac{1}{2}0$; the coordinates of the S atoms are $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; $\frac{1}{4}\frac{3}{4}\frac{3}{4}$; $\frac{3}{4}\frac{1}{4}\frac{3}{4}$; $\frac{3}{4}\frac{3}{4}\frac{1}{4}$. The lattice is fcc. There are four molecules of ZnS per conventional cell. About each atom there are four equally distant atoms of the opposite kind arranged at the corners of a regular tetrahedron.

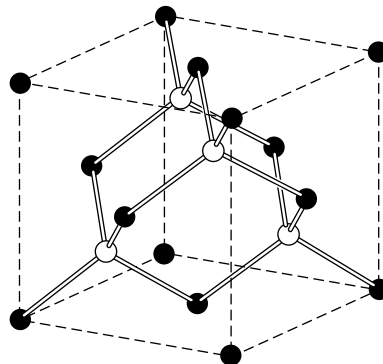


Figure 24 Crystal structure of cubic zinc sulfide.

The diamond structure allows a center-of-inversion symmetry operation at the midpoint of every line between nearest-neighbor atoms. The inversion operation carries an atom at $\mathbf{r}$ into an atom at $-\mathbf{r}$. The cubic ZnS structure does not have inversion symmetry. Examples of the cubic zinc sulfide structure are

Crystal	a	Crystal	a
SiC	4.35 Å	ZnSe	5.65 Å
ZnS	5.41	GaAs	5.65
AlP	5.45	AlAs	5.66
GaP	5.45	InSb	6.46

The close equality of the lattice constants of several pairs, notably (Al, Ga)P and (Al, Ga)As, makes possible the construction of semiconductor heterojunctions (Chapter 19).

DIRECT IMAGING OF ATOMIC STRUCTURE

Direct images of crystal structure have been produced by transmission electron microscopy. Perhaps the most beautiful images are produced by scanning tunneling microscopy; in STM (Chapter 19) one exploits the large variations in quantum tunneling as a function of the height of a fine metal tip above the surface of a crystal. The image of Fig. 25 was produced in this way. An STM method has been developed that will assemble single atoms into an organized layer nanometer structure on a crystal substrate.

NONIDEAL CRYSTAL STRUCTURES

The ideal crystal of classical crystallographers is formed by the periodic repetition of identical units in space. But no general proof has been given that

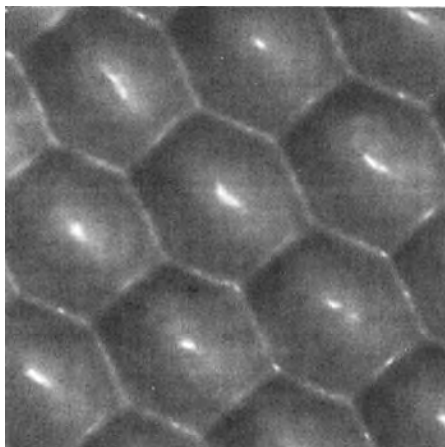


Figure 25 A scanning tunneling microscope image of atoms on a (111) surface of fcc platinum at 4 K. The nearest-neighbor spacing is 2.78 Å. (Photo courtesy of D. M. Eigler, IBM Research Division.)

the ideal crystal is the state of minimum energy of identical atoms at the temperature of absolute zero. At finite temperatures this is likely not to be true. We give a further example here.

Random Stacking and Polytypism

The fcc and hcp structures are made up of close-packed planes of atoms. The structures differ in the stacking sequence of the planes, fcc having the sequence $ABCABC \dots$ and hcp having the sequence $ABABAB \dots$. Structures are known in which the stacking sequence of close-packed planes is random. This is known as **random stacking** and may be thought of as crystalline in two dimensions and noncrystalline or glasslike in the third.

Polytypism is characterized by a stacking sequence with a long repeat unit along the stacking axis. The best known example is zinc sulfide, ZnS, in which more than 150 polytypes have been identified, with the longest periodicity being 360 layers. Another example is silicon carbide, SiC, which occurs with more than 45 stacking sequences of the close-packed layers. The polytype of SiC known as 393R has a primitive cell with $a = 3.079 \text{ Å}$ and $c = 989.6 \text{ Å}$. The longest primitive cell observed for SiC has a repeat distance of 594 layers. A given sequence is repeated many times within a single crystal. The mechanism that induces such long-range crystallographic order is not a long-range force, but arises from spiral steps due to dislocations in the growth nucleus (Chapter 20).

CRYSTAL STRUCTURE DATA

In Table 3 we list the more common crystal structures and lattice structures of the elements. Values of the atomic concentration and the density are given in Table 4. Many elements occur in several crystal structures and transform from

one to the other as the temperature or pressure is varied. Sometimes two structures coexist at the same temperature and pressure, although one may be slightly more stable.

SUMMARY

- A lattice is an array of points related by the lattice translation operator $\mathbf{T} = u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + u_3\mathbf{a}_3$, where u_1, u_2, u_3 are integers and $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are the crystal axes.
- To form a crystal we attach to every lattice point an identical basis composed of s atoms at the positions $\mathbf{r}_j = x_j\mathbf{a}_1 + y_j\mathbf{a}_2 + z_j\mathbf{a}_3$, with $j = 1, 2, \dots, s$. Here x, y, z may be selected to have values between 0 and 1.
- The axes $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are primitive for the minimum cell volume $|\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3|$ for which the crystal can be constructed from a lattice translation operator $\mathbf{T}$ and a basis at every lattice point.

Problems

1. **Tetrahedral angles.** The angles between the tetrahedral bonds of diamond are the same as the angles between the body diagonals of a cube, as in Fig. 10. Use elementary vector analysis to find the value of the angle.
2. **Indices of planes.** Consider the planes with indices (100) and (001); the lattice is fcc, and the indices refer to the conventional cubic cell. What are the indices of these planes when referred to the primitive axes of Fig. 11?
3. **Hcp structure.** Show that the c/a ratio for an ideal hexagonal close-packed structure is $(\frac{8}{3})^{1/2} = 1.633$. If c/a is significantly larger than this value, the crystal structure may be thought of as composed of planes of closely packed atoms, the planes being loosely stacked.

2

Wave Diffraction and the Reciprocal Lattice

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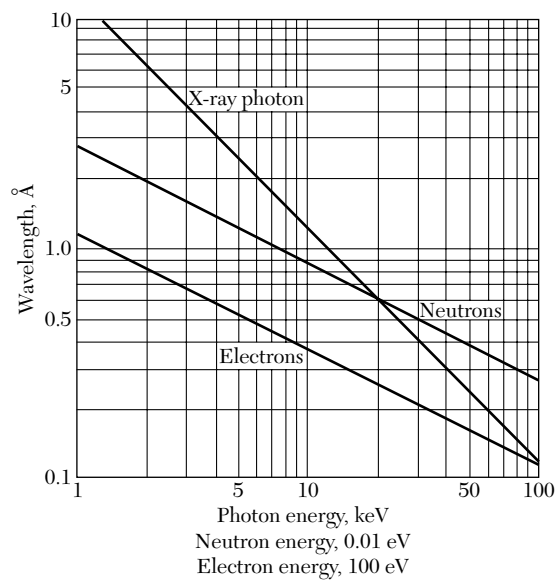


Figure 1 Wavelength versus particle energy, for photons, neutrons, and electrons.

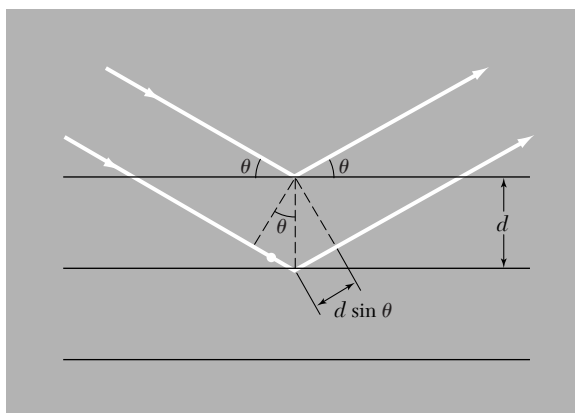


Figure 2 Derivation of the Bragg equation $2d \sin \theta = n\lambda$; here d is the spacing of parallel atomic planes and $2\pi n$ is the difference in phase between reflections from successive planes. The reflecting planes have nothing to do with the surface planes bounding the particular specimen.

CHAPTER 2: WAVE DIFFRACTION AND THE RECIPROCAL LATTICE

DIFFRACTION OF WAVES BY CRYSTALS

The Bragg law

We study crystal structure through the diffraction of photons, neutrons, and electrons (Fig. 1). The diffraction depends on the crystal structure and on the wavelength. At optical wavelengths such as 5000 Å, the superposition of the waves scattered elastically by the individual atoms of a crystal results in ordinary optical refraction. When the wavelength of the radiation is comparable with or smaller than the lattice constant, we may find diffracted beams in directions quite different from the incident direction.

W. L. Bragg presented a simple explanation of the diffracted beams from a crystal. The Bragg derivation is simple but is convincing only because it reproduces the correct result. Suppose that the incident waves are reflected specularly from parallel planes of atoms in the crystal, with each plane reflecting only a very small fraction of the radiation, like a lightly silvered mirror. In specular (mirrorlike) reflection the angle of incidence is equal to the angle of reflection. The diffracted beams are found when the reflections from parallel planes of atoms interfere constructively, as in Fig. 2. We treat elastic scattering, in which the energy of the x-ray is not changed on reflection.

Consider parallel lattice planes spaced d apart. The radiation is incident in the plane of the paper. The path difference for rays reflected from adjacent planes is $2d \sin \theta$, where θ is measured from the plane. Constructive interference of the radiation from successive planes occurs when the path difference is an integral number n of wavelengths λ , so that

$$2d \sin \theta = n\lambda . \quad (1)$$

This is the Bragg law, which can be satisfied only for wavelength $\lambda \leq 2d$.

Although the reflection from each plane is specular, for only certain values of θ will the reflections from all periodic parallel planes add up in phase to give a strong reflected beam. If each plane were perfectly reflecting, only the first plane of a parallel set would see the radiation, and any wavelength would be reflected. But each plane reflects 10^{-3} to 10^{-5} of the incident radiation, so that 10^3 to 10^5 planes may contribute to the formation of the Bragg-reflected beam in a perfect crystal. Reflection by a single plane of atoms is treated in Chapter 17 on surface physics.

The Bragg law is a consequence of the periodicity of the lattice. Notice that the law does not refer to the composition of the basis of atoms associated

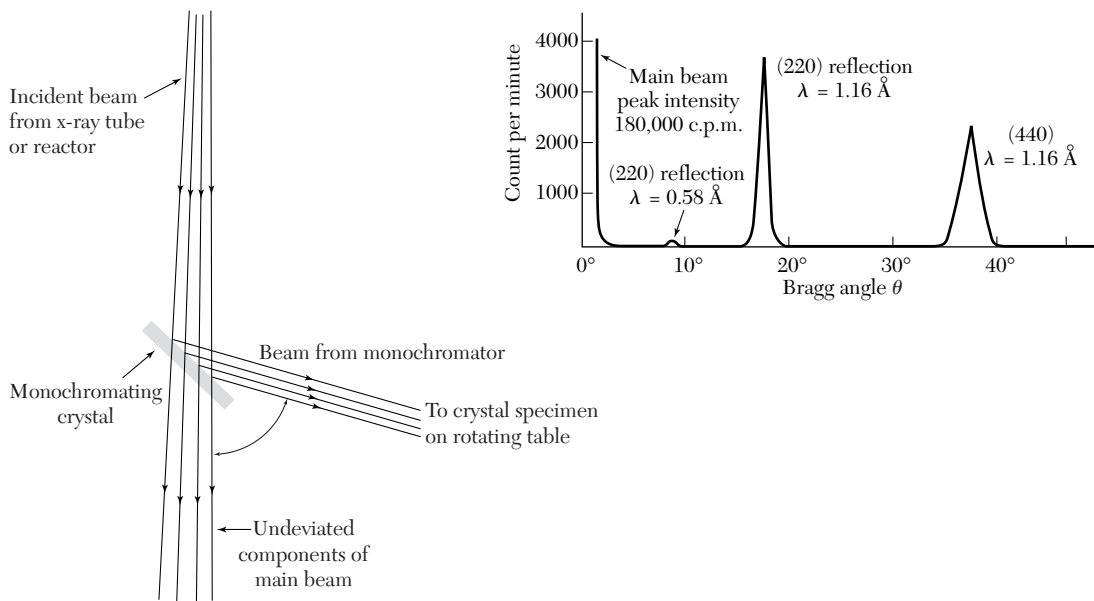


Figure 3 Sketch of a monochromator which by Bragg reflection selects a narrow spectrum of x-ray or neutron wavelengths from a broad spectrum incident beam. The upper part of the figure shows the analysis (obtained by reflection from a second crystal) of the purity of a 1.16 Å beam of neutrons from a calcium fluoride crystal monochromator. (After G. Bacon.)

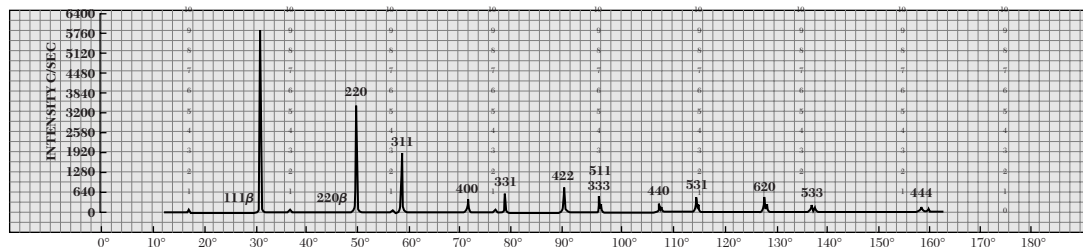


Figure 4 X-ray diffractometer recording of powdered silicon, showing a counter recording of the diffracted beams. (Courtesy of W. Parrish.)

with every lattice point. We shall see, however, that the composition of the basis determines the relative intensity of the various orders of diffraction (denoted by n above) from a given set of parallel planes. Bragg reflection from a single crystal is shown in Fig. 3 and from a powder in Fig. 4.

SCATTERED WAVE AMPLITUDE

The Bragg derivation of the diffraction condition (1) gives a neat statement of the condition for the constructive interference of waves scattered from the lattice points. We need a deeper analysis to determine the scattering

intensity from the basis of atoms, which means from the spatial distribution of electrons within each cell.

Fourier Analysis

We have seen that a crystal is invariant under any translation of the form $\mathbf{T} = u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + u_3\mathbf{a}_3$, where u_1, u_2, u_3 are integers and $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are the crystal axes. Any local physical property of the crystal, such as the charge concentration, electron number density, or magnetic moment density is invariant under $\mathbf{T}$. What is most important to us here is that the electron number density $n(\mathbf{r})$ is a periodic function of $\mathbf{r}$, with periods $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ in the directions of the three crystal axes, respectively. Thus

$$n(\mathbf{r} + \mathbf{T}) = n(\mathbf{r}) . \quad (2)$$

Such periodicity creates an ideal situation for Fourier analysis. The most interesting properties of crystals are directly related to the Fourier components of the electron density.

We consider first a function $n(x)$ in one dimension with period a in the direction x . We expand $n(x)$ in a Fourier series of sines and cosines:

$$n(x) = n_0 + \sum_{p>0} [C_p \cos(2\pi p x/a) + S_p \sin(2\pi p x/a)] , \quad (3)$$

where the p are positive integers and C_p, S_p are real constants, called the Fourier coefficients of the expansion. The factor $2\pi/a$ in the arguments ensures that $n(x)$ has the period a :

$$\begin{aligned} n(x+a) &= n_0 + \sum [C_p \cos(2\pi p x/a + 2\pi p) + S_p \sin(2\pi p x/a + 2\pi p)] \\ &= n_0 + \sum [C_p \cos(2\pi p x/a) + S_p \sin(2\pi p x/a)] = n(x) . \end{aligned} \quad (4)$$

We say that $2\pi p/a$ is a point in the reciprocal lattice or Fourier space of the crystal. In one dimension these points lie on a line. The **reciprocal lattice points** tell us the allowed terms in the Fourier series (4) or (5). A term is allowed if it is consistent with the periodicity of the crystal, as in Fig. 5; other

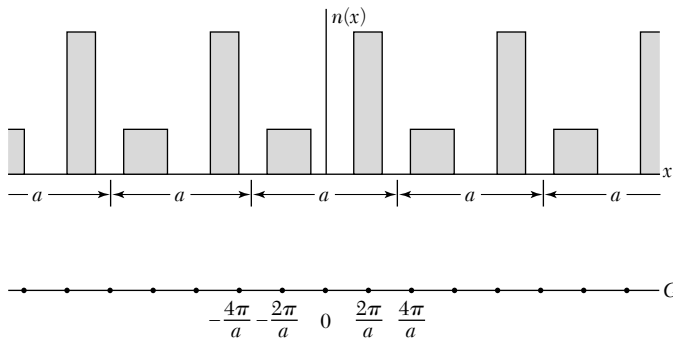


Figure 5 A periodic function $n(x)$ of period a , and the terms $2\pi p/a$ that may appear in the Fourier transform $n(x) = \sum n_p \exp(i2\pi p x/a)$.

points in the reciprocal space are not allowed in the Fourier expansion of a periodic function.

It is convenient to write the series (4) in the compact form

$$n(x) = \sum_p n_p \exp(i2\pi px/a) , \quad (5)$$

where the sum is over all integers p : positive, negative, and zero. The coefficients n_p now are complex numbers. To ensure that $n(x)$ is a real function, we require

$$n_{-p}^* = n_p , \quad (6)$$

for then the sum of the terms in p and $-p$ is real. The asterisk on n_{-p}^* denotes the complex conjugate of n_{-p} .

With $\varphi = 2\pi px/a$, the sum of the terms in p and $-p$ in (5) is real if (6) is satisfied. The sum is

$$\begin{aligned} n_p(\cos \varphi + i \sin \varphi) + n_{-p}(\cos \varphi - i \sin \varphi) \\ = (n_p + n_{-p})\cos \varphi + i(n_p - n_{-p})\sin \varphi , \end{aligned} \quad (7)$$

which in turn is equal to the real function

$$2\text{Re}\{n_p\} \cos \varphi - 2\text{Im}\{n_p\} \sin \varphi \quad (8)$$

if (6) is satisfied. Here $\text{Re}\{n_p\}$ and $\text{Im}\{n_p\}$ are real and denote the real and imaginary parts of n_p . Thus the number density $n(x)$ is a real function, as desired.

The extension of the Fourier analysis to periodic functions $n(\mathbf{r})$ in three dimensions is straightforward. We must find a set of vectors $\mathbf{G}$ such that

$$n(\mathbf{r}) = \sum_{\mathbf{G}} n_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r}) \quad (9)$$

is invariant under all crystal translations $\mathbf{T}$ that leave the crystal invariant. It will be shown below that the set of Fourier coefficients $n_{\mathbf{G}}$ determines the x-ray scattering amplitude.

Inversion of Fourier Series. We now show that the Fourier coefficient n_p in the series (5) is given by

$$n_p = a^{-1} \int_0^a dx n(x) \exp(-i2\pi px/a) . \quad (10)$$

Substitute (5) in (10) to obtain

$$n_p = a^{-1} \sum_{p'} n_{p'} \int_0^a dx \exp[i2\pi(p' - p)x/a] . \quad (11)$$

If $p' \neq p$ the value of the integral is

$$\frac{a}{i2\pi(p' - p)} (e^{i2\pi(p' - p)} - 1) = 0 ,$$

because $p' - p$ is an integer and $\exp[i2\pi(\text{integer})] = 1$. For the term $p' = p$ the integrand is $\exp(i0) = 1$, and the value of the integral is a , so that $n_p = a^{-1}n_p a = n_p$, which is an identity, so that (10) is an identity.

As in (10), the inversion of (9) gives

$$n_{\mathbf{G}} = V_c^{-1} \int_{\text{cell}} dV n(\mathbf{r}) \exp(-i\mathbf{G} \cdot \mathbf{r}) . \quad (12)$$

Here V_c is the volume of a cell of the crystal.

Reciprocal Lattice Vectors

To proceed further with the Fourier analysis of the electron concentration we must find the vectors $\mathbf{G}$ of the Fourier sum $\sum n_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$ as in (9). There is a powerful, somewhat abstract procedure for doing this. The procedure forms the theoretical basis for much of solid state physics, where Fourier analysis is the order of the day.

We construct the axis vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ of the **reciprocal lattice**:

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} ; \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} ; \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} . \quad (13)$$

The factors 2π are not used by crystallographers but are convenient in solid state physics.

If $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are primitive vectors of the crystal lattice, then $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are **primitive vectors of the reciprocal lattice**. Each vector defined by (13) is orthogonal to two axis vectors of the crystal lattice. Thus $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ have the property

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij} , \quad (14)$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

Points in the reciprocal lattice are mapped by the set of vectors

$$\mathbf{G} = v_1 \mathbf{b}_1 + v_2 \mathbf{b}_2 + v_3 \mathbf{b}_3 , \quad (15)$$

where v_1, v_2, v_3 are integers. A vector $\mathbf{G}$ of this form is a **reciprocal lattice** vector.

The vectors $\mathbf{G}$ in the Fourier series (9) are just the reciprocal lattice vectors (15), for then the Fourier series representation of the electron density has the desired invariance under any crystal translation $\mathbf{T} = u_1 \mathbf{a}_1 + u_2 \mathbf{a}_2 + u_3 \mathbf{a}_3$. From (9),

$$n(\mathbf{r} + \mathbf{T}) = \sum_{\mathbf{G}} n_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r}) \exp(i\mathbf{G} \cdot \mathbf{T}) . \quad (16)$$

But $\exp(i\mathbf{G} \cdot \mathbf{T}) = 1$, because

$$\begin{aligned}\exp(i\mathbf{G} \cdot \mathbf{T}) &= \exp[i(v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + v_3\mathbf{b}_3) \cdot (u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + u_3\mathbf{a}_3)] \\ &= \exp[i2\pi(v_1u_1 + v_2u_2 + v_3u_3)] .\end{aligned}\quad (17)$$

The argument of the exponential has the form $2\pi i$ times an integer, because $v_1u_1 + v_2u_2 + v_3u_3$ is an integer, being the sum of products of integers. Thus by (9) we have the desired invariance, $n(\mathbf{r} + \mathbf{T}) = n(\mathbf{r}) = \sum n_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$.

Every crystal structure has two lattices associated with it, the crystal lattice and the reciprocal lattice. A diffraction pattern of a crystal is, as we shall show, a map of the reciprocal lattice of the crystal. A microscope image, if it could be resolved on a fine enough scale, is a map of the crystal structure in real space. The two lattices are related by the definitions (13). Thus when we rotate a crystal in a holder, we rotate both the direct lattice and the reciprocal lattice.

Vectors in the direct lattice have the dimensions of [length]; vectors in the reciprocal lattice have the dimensions of [1/length]. The reciprocal lattice is a lattice in the Fourier space associated with the crystal. The term is motivated below. Wavevectors are always drawn in Fourier space, so that every position in Fourier space may have a meaning as a description of a wave, but there is a special significance to the points defined by the set of $\mathbf{G}$'s associated with a crystal structure.

Diffraction Conditions

Theorem. The set of reciprocal lattice vectors $\mathbf{G}$ determines the possible x-ray reflections.

We see in Fig. 6 that the difference in phase factors is $\exp[i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}]$ between beams scattered from volume elements $\mathbf{r}$ apart. The wavevectors of the incoming and outgoing beams are $\mathbf{k}$ and $\mathbf{k}'$. We suppose that the amplitude

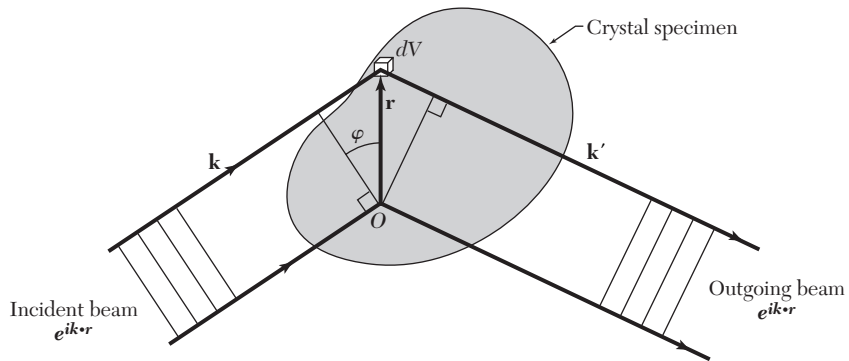


Figure 6 The difference in path length of the incident wave $\mathbf{k}$ at the points O , $\mathbf{r}$ is $r \sin \phi$, and the difference in phase angle is $(2\pi r \sin \phi) / \lambda$, which is equal to $\mathbf{k} \cdot \mathbf{r}$. For the diffracted wave the difference in phase angle is $-\mathbf{k}' \cdot \mathbf{r}$. The total difference in phase angle is $(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}$, and the wave scattered from dV at $\mathbf{r}$ has the phase factor $\exp[i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}]$ relative to the wave scattered from a volume element at the origin O .

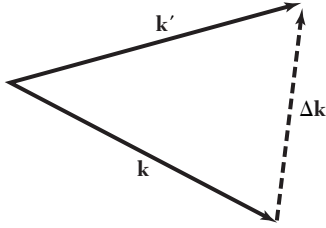


Figure 7 Definition of the scattering vector $\Delta\mathbf{k}$ such that $\mathbf{k} + \Delta\mathbf{k} = \mathbf{k}'$. In elastic scattering the magnitudes satisfy $k' = k$. Further, in Bragg scattering from a periodic lattice, any allowed $\Delta\mathbf{k}$ must equal some reciprocal lattice vector $\mathbf{G}$.

of the wave scattered from a volume element is proportional to the local electron concentration $n(\mathbf{r})$. The total amplitude of the scattered wave in the direction of $\mathbf{k}'$ is proportional to the integral over the crystal of $n(\mathbf{r}) dV$ times the phase factor $\exp[i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}]$.

In other words, the amplitude of the electric or magnetic field vectors in the scattered electromagnetic wave is proportional to the following integral which defines the quantity F that we call the **scattering amplitude**:

$$F = \int dV n(\mathbf{r}) \exp[i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}] = \int dV n(\mathbf{r}) \exp(-i\Delta\mathbf{k} \cdot \mathbf{r}) , \quad (18)$$

where $\mathbf{k} - \mathbf{k}' = -\Delta\mathbf{k}$, or

$$\mathbf{k} + \Delta\mathbf{k} = \mathbf{k}' . \quad (19)$$

Here $\Delta\mathbf{k}$ measures the change in wavevector and is called the **scattering vector** (Fig. 7). We add $\Delta\mathbf{k}$ to $\mathbf{k}$ to obtain $\mathbf{k}'$, the wavevector of the scattered beam.

We introduce into (18) the Fourier components (9) of $n(\mathbf{r})$ to obtain for the scattering amplitude

$$F = \sum_{\mathbf{G}} \int dV n_{\mathbf{G}} \exp[i(\mathbf{G} - \Delta\mathbf{k}) \cdot \mathbf{r}] . \quad (20)$$

When the scattering vector $\Delta\mathbf{k}$ is equal to a particular reciprocal lattice vector,

$$\Delta\mathbf{k} = \mathbf{G} , \quad (21)$$

the argument of the exponential vanishes and $F = Vn_{\mathbf{G}}$. It is a simple exercise (Problem 4) to show that F is negligibly small when $\Delta\mathbf{k}$ differs significantly from any reciprocal lattice vector.

In elastic scattering of a photon its energy $\hbar\omega$ is conserved, so that the frequency $\omega' = ck'$ of the emergent beam is equal to the frequency of the incident beam. Thus the magnitudes k and k' are equal, and $k^2 = k'^2$, a result that holds also for elastic scattering of electron and neutron beams. From (21) we found $\Delta\mathbf{k} = \mathbf{G}$ or $\mathbf{k} + \mathbf{G} = \mathbf{k}'$, so that the **diffraction condition** is written as $(\mathbf{k} + \mathbf{G})^2 = k^2$, or

$$2\mathbf{k} \cdot \mathbf{G} + G^2 = 0 . \quad (22)$$

This is the central result of the theory of elastic scattering of waves in a periodic lattice. If $\mathbf{G}$ is a reciprocal lattice vector, so is $-\mathbf{G}$, and with this substitution we can write (22) as

$$2\mathbf{k} \cdot \mathbf{G} = G^2 . \quad (23)$$

This particular expression is often used as the condition for diffraction.

Equation (23) is another statement of the Bragg condition (1). The result of Problem 1 is that the spacing $d(hkl)$ between parallel lattice planes that are normal to the direction $\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ is $d(hkl) = 2\pi/|\mathbf{G}|$. Thus the result $2\mathbf{k} \cdot \mathbf{G} = G^2$ may be written as

$$2(2\pi/\lambda) \sin \theta = 2\pi/d(hkl) ,$$

or $2d(hkl) \sin \theta = \lambda$. Here θ is the angle between the incident beam and the crystal plane.

The integers hkl that define $\mathbf{G}$ are not necessarily identical with the indices of an actual crystal plane, because the hkl may contain a common factor n , whereas in the definition of the indices in Chapter 1 the common factor has been eliminated. We thus obtain the Bragg result:

$$2d \sin \theta = n\lambda , \quad (24)$$

where d is the spacing between adjacent parallel planes with indices h/n , k/n , l/n .

Laue Equations

The original result (21) of diffraction theory, namely that $\Delta\mathbf{k} = \mathbf{G}$, may be expressed in another way to give what are called the Laue equations. These are valuable because of their geometrical representation. Take the scalar product of both $\Delta\mathbf{k}$ and $\mathbf{G}$ successively with $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$. From (14) and (15) we get

$$\mathbf{a}_1 \cdot \Delta\mathbf{k} = 2\pi v_1 ; \quad \mathbf{a}_2 \cdot \Delta\mathbf{k} = 2\pi v_2 ; \quad \mathbf{a}_3 \cdot \Delta\mathbf{k} = 2\pi v_3 . \quad (25)$$

These equations have a simple geometrical interpretation. The first equation $\mathbf{a}_1 \cdot \Delta\mathbf{k} = 2\pi v_1$ tells us that $\Delta\mathbf{k}$ lies on a certain cone about the direction of $\mathbf{a}_1$. The second equation tells us that $\Delta\mathbf{k}$ lies on a cone about $\mathbf{a}_2$ as well, and the third equation requires that $\Delta\mathbf{k}$ lies on a cone about $\mathbf{a}_3$. Thus, at a reflection $\Delta\mathbf{k}$ must satisfy all three equations; it must lie at the common line of intersection of *three* cones, which is a severe condition that can be satisfied only by systematic sweeping or searching in wavelength or crystal orientation—or by sheer accident.

A beautiful construction, the Ewald construction, is exhibited in Fig. 8. This helps us visualize the nature of the accident that must occur in order to satisfy the diffraction condition in three dimensions.

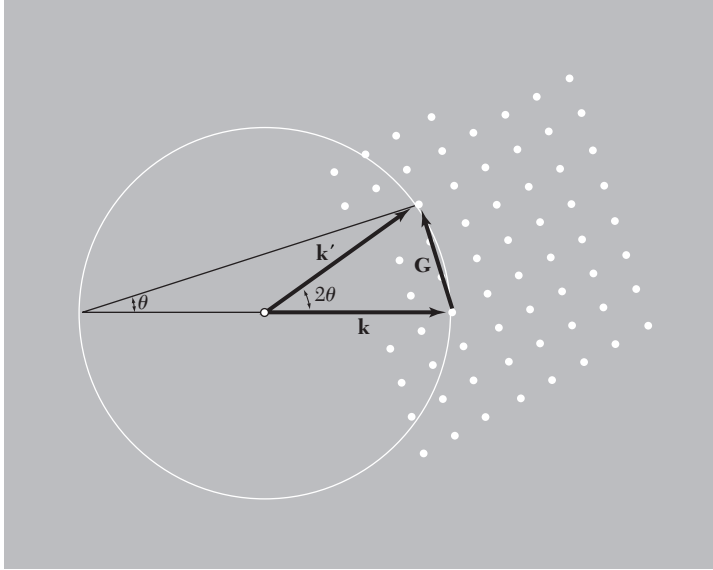


Figure 8 The points on the right-hand side are reciprocal-lattice points of the crystal. The vector $\mathbf{k}$ is drawn in the direction of the incident x-ray beam, and the origin is chosen such that $\mathbf{k}$ terminates at any reciprocal lattice point. We draw a sphere of radius $k = 2\pi/\lambda$ about the origin of $\mathbf{k}$. A diffracted beam will be formed if this sphere intersects any other point in the reciprocal lattice. The sphere as drawn intercepts a point connected with the end of $\mathbf{k}$ by a reciprocal lattice vector $\mathbf{G}$. The diffracted x-ray beam is in the direction $\mathbf{k}' = \mathbf{k} + \mathbf{G}$. The angle θ is the Bragg angle of Fig. 2. This construction is due to P. P. Ewald.

BRILLOUIN ZONES

Brillouin gave the statement of the diffraction condition that is most widely used in solid state physics, which means in the description of electron energy band theory and of the elementary excitations of other kinds. A Brillouin zone is defined as a Wigner-Seitz primitive cell in the reciprocal lattice. (The construction in the direct lattice was shown in Fig. 1.4.) The Brillouin zone gives a vivid geometrical interpretation of the diffraction condition $2\mathbf{k} \cdot \mathbf{G} = G^2$ of Eq. (23). We divide both sides by 4 to obtain

$$\mathbf{k} \cdot \left(\frac{1}{2}\mathbf{G}\right) = \left(\frac{1}{2}G\right)^2. \quad (26)$$

We now work in reciprocal space, the space of the $\mathbf{k}$'s and $\mathbf{G}$'s. Select a vector $\mathbf{G}$ from the origin to a reciprocal lattice point. Construct a plane normal to this vector $\mathbf{G}$ at its midpoint. This plane forms a part of a zone boundary (Fig. 9a). An x-ray beam in the crystal will be diffracted if its wavevector $\mathbf{k}$ has the magnitude and direction required by (26). The diffracted beam will then be in the direction $\mathbf{k} - \mathbf{G}$, as we see from (19) with $\Delta\mathbf{k} = -\mathbf{G}$. Thus the Brillouin construction exhibits all the wavevectors $\mathbf{k}$ which can be Bragg-reflected by the crystal.

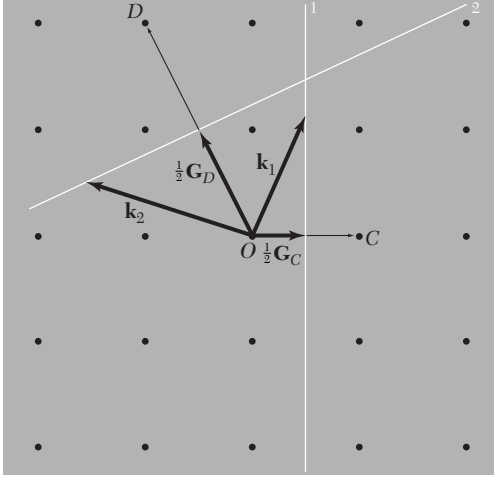


Figure 9a Reciprocal lattice points near the point O at the origin of the reciprocal lattice. The reciprocal lattice vector $\mathbf{G}_C$ connects points OC ; and $\mathbf{G}_D$ connects OD . Two planes 1 and 2 are drawn which are the perpendicular bisectors of $\mathbf{G}_C$ and $\mathbf{G}_D$, respectively. Any vector from the origin to the plane 1, such as $\mathbf{k}_1$, will satisfy the diffraction condition $\mathbf{k}_1 \cdot (\frac{1}{2} \mathbf{G}_C) = (\frac{1}{2} \mathbf{G}_C)^2$. Any vector from the origin to the plane 2, such as $\mathbf{k}_2$, will satisfy the diffraction condition $\mathbf{k}_2 \cdot (\frac{1}{2} \mathbf{G}_D) = (\frac{1}{2} \mathbf{G}_D)^2$.

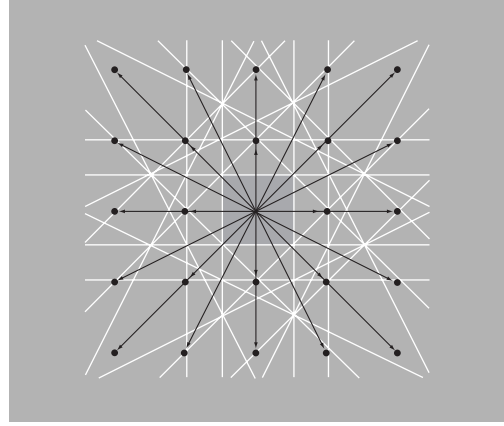


Figure 9b Square reciprocal lattice with reciprocal lattice vectors shown as fine black lines. The lines shown in white are perpendicular bisectors of the reciprocal lattice vectors. The central square is the smallest volume about the origin which is bounded entirely by white lines. The square is the Wigner-Seitz primitive cell of the reciprocal lattice. It is called the first Brillouin zone.

The set of planes that are the perpendicular bisectors of the reciprocal lattice vectors is of general importance in the theory of wave propagation in crystals: A wave whose wavevector drawn from the origin terminates on any of these planes will satisfy the condition for diffraction. These planes divide the Fourier space of the crystal into fragments, as shown in Fig. 9b for a square lattice. The central square is a primitive cell of the reciprocal lattice. It is a Wigner-Seitz cell of the reciprocal lattice.

The central cell in the reciprocal lattice is of special importance in the theory of solids, and we call it the first Brillouin zone. *The first Brillouin zone is the smallest volume entirely enclosed by planes that are the perpendicular bisectors of the reciprocal lattice vectors drawn from the origin.* Examples are shown in Figs. 10 and 11.

Historically, Brillouin zones are not part of the language of x-ray diffraction analysis of crystal structures, but the zones are an essential part of the analysis of the electronic energy-band structure of crystals.

Reciprocal Lattice to sc Lattice

The primitive translation vectors of a simple cubic lattice may be taken as the set

$$\mathbf{a}_1 = a\hat{\mathbf{x}} ; \quad \mathbf{a}_2 = a\hat{\mathbf{y}} ; \quad \mathbf{a}_3 = a\hat{\mathbf{z}} . \quad (27a)$$

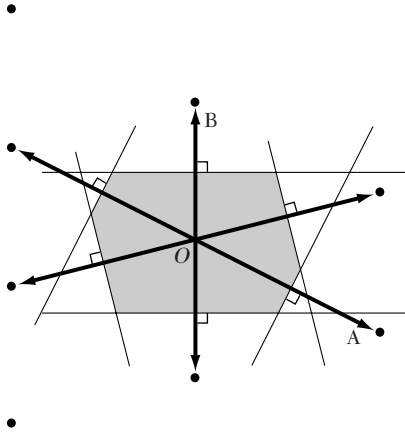


Figure 10 Construction of the first Brillouin zone for an oblique lattice in two dimensions. We first draw a number of vectors from O to nearby points in the reciprocal lattice. Next we construct lines perpendicular to these vectors at their mid-points. The smallest enclosed area is the first Brillouin zone.

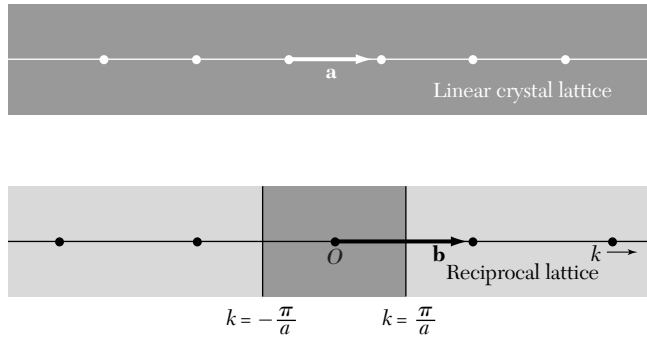


Figure 11 Crystal and reciprocal lattices in one dimension. The basis vector in the reciprocal lattice is $\mathbf{b}$, of length equal to $2\pi/a$. The shortest reciprocal lattice vectors from the origin are $\mathbf{b}$ and $-\mathbf{b}$. The perpendicular bisectors of these vectors form the boundaries of the first Brillouin zone. The boundaries are at $k = \pm\pi/a$.

Here $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ are orthogonal vectors of unit length. The volume of the cell is $\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3 = a^3$. The primitive translation vectors of the reciprocal lattice are found from the standard prescription (13):

$$\mathbf{b}_1 = (2\pi/a)\hat{\mathbf{x}} ; \quad \mathbf{b}_2 = (2\pi/a)\hat{\mathbf{y}} ; \quad \mathbf{b}_3 = (2\pi/a)\hat{\mathbf{z}} . \quad (27b)$$

Here the reciprocal lattice is itself a simple cubic lattice, now of lattice constant $2\pi/a$.

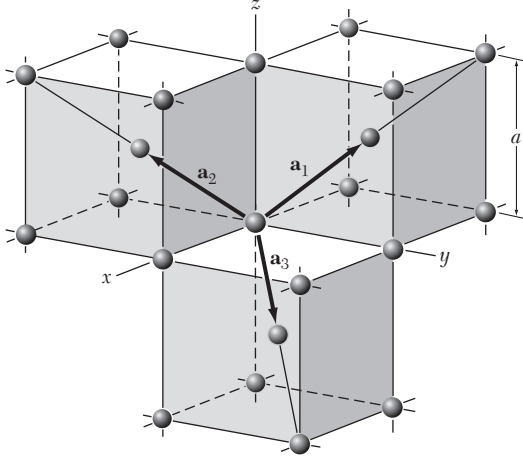


Figure 12 Primitive basis vectors of the body-centered cubic lattice.

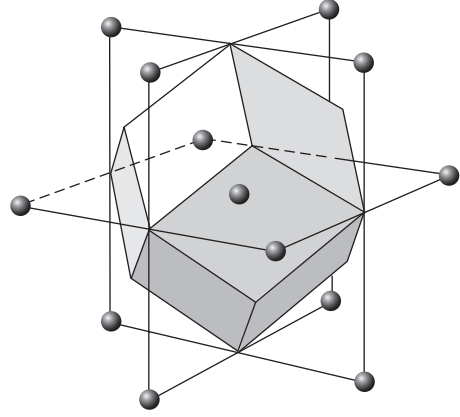


Figure 13 First Brillouin zone of the body-centered cubic lattice. The figure is a regular rhombic dodecahedron.

The boundaries of the first Brillouin zones are the planes normal to the six reciprocal lattice vectors $\pm \mathbf{b}_1, \pm \mathbf{b}_2, \pm \mathbf{b}_3$ at their midpoints:

$$\pm \frac{1}{2} \mathbf{b}_1 = \pm (\pi/a) \hat{\mathbf{x}} ; \quad \pm \frac{1}{2} \mathbf{b}_2 = \pm (\pi/a) \hat{\mathbf{y}} ; \quad \pm \frac{1}{2} \mathbf{b}_3 = \pm (\pi/a) \hat{\mathbf{z}} . \quad (28)$$

The six planes bound a cube of edge $2\pi/a$ and of volume $(2\pi/a)^3$; this cube is the first Brillouin zone of the sc crystal lattice.

Reciprocal Lattice to bcc Lattice

The primitive translation vectors of the bcc lattice (Fig. 12) are

$$\mathbf{a}_1 = \frac{1}{2}a(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) ; \quad \mathbf{a}_2 = \frac{1}{2}a(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}) ; \quad \mathbf{a}_3 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}) , \quad (29)$$

where a is the side of the conventional cube and $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ are orthogonal unit vectors parallel to the cube edges. The volume of the primitive cell is

$$V = |\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3| = \frac{1}{2}a^3 . \quad (30)$$

The primitive translations of the reciprocal lattice are defined by (13). We have, using (28),

$$\mathbf{b}_1 = (2\pi/a)(\hat{\mathbf{y}} + \hat{\mathbf{z}}) ; \quad \mathbf{b}_2 = (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{z}}) ; \quad \mathbf{b}_3 = (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{y}}) . \quad (31)$$

Note by comparison with Fig. 14 (p. 37) that these are just the primitive vectors of an fcc lattice, so that an fcc lattice is the reciprocal lattice of the bcc lattice.

The general reciprocal lattice vector is, for integral v_1, v_2, v_3 ,

$$\mathbf{G} = v_1 \mathbf{b}_1 + v_2 \mathbf{b}_2 + v_3 \mathbf{b}_3 = (2\pi/a)[(v_2 + v_3)\hat{\mathbf{x}} + (v_1 + v_3)\hat{\mathbf{y}} + (v_1 + v_2)\hat{\mathbf{z}}] . \quad (32)$$

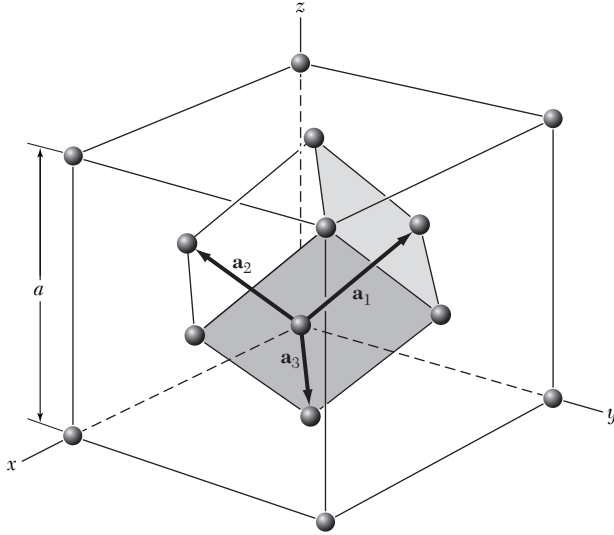


Figure 14 Primitive basis vectors of the face-centered cubic lattice.

The shortest $\mathbf{G}$'s are the following 12 vectors, where all choices of sign are independent:

$$(2\pi/a)(\pm\hat{y} \pm \hat{z}) ; \quad (2\pi/a)(\pm\hat{x} \pm \hat{z}) ; \quad (2\pi/a)(\pm\hat{x} \pm \hat{y}) . \quad (33)$$

One primitive cell of the reciprocal lattice is the parallelepiped described by the $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$ defined by (31). The volume of this cell in reciprocal space is $\mathbf{b}_1 \cdot \mathbf{b}_2 \times \mathbf{b}_3 = 2(2\pi/a)^3$. The cell contains one reciprocal lattice point, because each of the eight corner points is shared among eight parallelepipeds. Each parallelepiped contains one-eighth of each of eight corner points (see Fig. 12).

Another primitive cell is the central (Wigner-Seitz) cell of the reciprocal lattice which is the first Brillouin zone. Each such cell contains one lattice point at the central point of the cell. This zone (for the bcc lattice) is bounded by the planes normal to the 12 vectors of Eq. (33) at their midpoints. The zone is a regular 12-faced solid, a rhombic dodecahedron, as shown in Fig. 13.

Reciprocal Lattice to fcc Lattice

The primitive translation vectors of the fcc lattice of Fig. 14 are

$$\mathbf{a}_1 = \frac{1}{2}a(\hat{y} + \hat{z}) ; \quad \mathbf{a}_2 = \frac{1}{2}a(\hat{x} + \hat{z}) ; \quad \mathbf{a}_3 = \frac{1}{2}a(\hat{x} + \hat{y}) . \quad (34)$$

The volume of the primitive cell is

$$V = |\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3| = \frac{1}{4}a^3 . \quad (35)$$

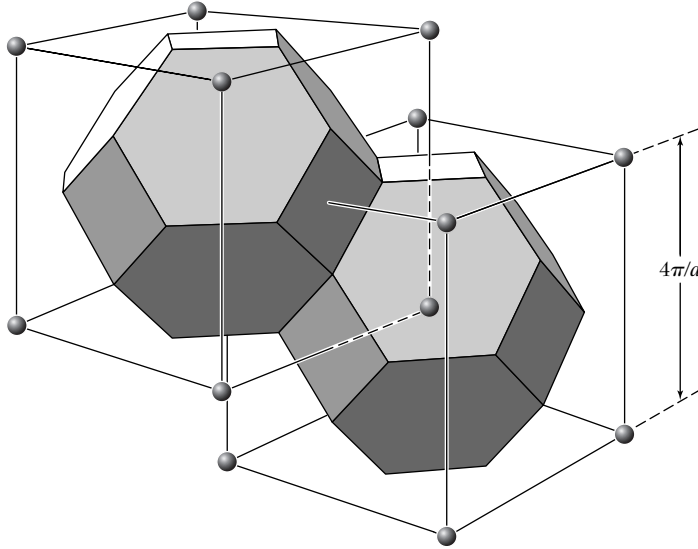


Figure 15 Brillouin zones of the face-centered cubic lattice. The cells are in reciprocal space, and the reciprocal lattice is body centered.

The primitive translation vectors of the lattice reciprocal to the fcc lattice are

$$\begin{aligned} \mathbf{b}_1 &= (2\pi/a)(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) ; & \mathbf{b}_2 &= (2\pi/a)(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}) ; \\ \mathbf{b}_3 &= (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}) . \end{aligned} \quad (36)$$

These are primitive translation vectors of a bcc lattice, so that the bcc lattice is reciprocal to the fcc lattice. The volume of the primitive cell of the reciprocal lattice is $4(2\pi/a)^3$.

The shortest $\mathbf{G}$'s are the eight vectors:

$$(2\pi/a)(\pm\hat{\mathbf{x}} \pm \hat{\mathbf{y}} \pm \hat{\mathbf{z}}) . \quad (37)$$

The boundaries of the central cell in the reciprocal lattice are determined for the most part by the eight planes normal to these vectors at their midpoints. But the corners of the octahedron thus formed are cut by the planes that are the perpendicular bisectors of six other reciprocal lattice vectors:

$$(2\pi/a)(\pm 2\hat{\mathbf{x}}) ; \quad (2\pi/a)(\pm 2\hat{\mathbf{y}}) ; \quad (2\pi/a)(\pm 2\hat{\mathbf{z}}) . \quad (38)$$

Note that $(2\pi/a)(2\hat{\mathbf{x}})$ is a reciprocal lattice vector because it is equal to $\mathbf{b}_2 + \mathbf{b}_3$. The first Brillouin zone is the smallest bounded volume about the origin, the truncated octahedron shown in Fig. 15. The six planes bound a cube of edge $4\pi/a$ and (before truncation) of volume $(4\pi/a)^3$.

FOURIER ANALYSIS OF THE BASIS

When the diffraction condition $\Delta \mathbf{k} = \mathbf{G}$ of Eq. (21) is satisfied, the scattering amplitude (18) for a crystal of N cells may be written as

$$F_{\mathbf{G}} = N \int_{\text{cell}} dV n(\mathbf{r}) \exp(-i\mathbf{G} \cdot \mathbf{r}) = NS_{\mathbf{G}} . \quad (39)$$

The quantity $S_{\mathbf{G}}$ is called the **structure factor** and is defined as an integral over a single cell, with $\mathbf{r} = 0$ at one corner.

Often it is useful to write the electron concentration $n(\mathbf{r})$ as the superposition of electron concentration functions n_j associated with each atom j of the cell. If $\mathbf{r}_j$ is the vector to the center of atom j , then the function $n_j(\mathbf{r} - \mathbf{r}_j)$ defines the contribution of that atom to the electron concentration at $\mathbf{r}$. The total electron concentration at $\mathbf{r}$ due to all atoms in the single cell is the sum

$$n(\mathbf{r}) = \sum_{j=1}^s n_j(\mathbf{r} - \mathbf{r}_j) \quad (40)$$

over the s atoms of the basis. The decomposition of $n(\mathbf{r})$ is not unique, for we cannot always say how much charge density is associated with each atom. This is not an important difficulty.

The structure factor defined by (39) may now be written as integrals over the s atoms of a cell:

$$\begin{aligned} S_{\mathbf{G}} &= \sum_j \int dV n_j(\mathbf{r} - \mathbf{r}_j) \exp(-i\mathbf{G} \cdot \mathbf{r}) \\ &= \sum_j \exp(-i\mathbf{G} \cdot \mathbf{r}_j) \int dV n_j(\boldsymbol{\rho}) \exp(-i\mathbf{G} \cdot \boldsymbol{\rho}) , \end{aligned} \quad (41)$$

where $\boldsymbol{\rho} = \mathbf{r} - \mathbf{r}_j$. We now define the **atomic form factor** as

$$f_j = \int dV n_j(\boldsymbol{\rho}) \exp(-i\mathbf{G} \cdot \boldsymbol{\rho}) , \quad (42)$$

integrated over all space. If $n_j(\boldsymbol{\rho})$ is an atomic property, f_j is an atomic property.

We combine (41) and (42) to obtain the **structure factor of the basis** in the form

$$S_{\mathbf{G}} = \sum_j f_j \exp(-i\mathbf{G} \cdot \mathbf{r}_j) . \quad (43)$$

The usual form of this result follows on writing for atom j :

$$\mathbf{r}_j = x_j \mathbf{a}_1 + y_j \mathbf{a}_2 + z_j \mathbf{a}_3 , \quad (44)$$

as in (1.2). Then, for the reflection labelled by v_1, v_2, v_3 , we have

$$\begin{aligned}\mathbf{G} \cdot \mathbf{r}_j &= (v_1 \mathbf{b}_1 + v_2 \mathbf{b}_2 + v_3 \mathbf{b}_3) \cdot (x_j \mathbf{a}_1 + y_j \mathbf{a}_2 + z_j \mathbf{a}_3) \\ &= 2\pi(v_1 x_j + v_2 y_j + v_3 z_j) \quad ,\end{aligned}\tag{45}$$

so that (43) becomes

$$S_{\mathbf{G}}(v_1 v_2 v_3) = \sum_j f_j \exp[-i2\pi(v_1 x_j + v_2 y_j + v_3 z_j)] \quad .\tag{46}$$

The structure factor S need not be real because the scattered intensity will involve S^*S , where S^* is the complex conjugate of S so that S^*S is real.

Structure Factor of the bcc Lattice

The bcc basis referred to the cubic cell has identical atoms at $x_1 = y_1 = z_1 = 0$ and at $x_2 = y_2 = z_2 = \frac{1}{2}$. Thus (46) becomes

$$S(v_1 v_2 v_3) = f\{1 + \exp[-i\pi(v_1 + v_2 + v_3)]\} \quad ,\tag{47}$$

where f is the form factor of an atom. The value of S is zero whenever the exponential has the value -1 , which is whenever the argument is $-i\pi \times (\text{odd integer})$. Thus we have

$$\begin{aligned}S &= 0 && \text{when } v_1 + v_2 + v_3 = \text{odd integer} \quad ; \\ S &= 2f && \text{when } v_1 + v_2 + v_3 = \text{even integer} \quad .\end{aligned}$$

Metallic sodium has a bcc structure. The diffraction pattern does not contain lines such as (100), (300), (111), or (221), but lines such as (200), (110), and (222) will be present; here the indices $(v_1 v_2 v_3)$ are referred to a cubic cell. What is the physical interpretation of the result that the (100) reflection vanishes? The (100) reflection normally occurs when reflections from the planes that bound the cubic cell differ in phase by 2π . In the bcc lattice there is an intervening plane (Fig. 16) of atoms, labeled the second plane in the figure, which is equal in scattering power to the other planes. Situated midway between them, it gives a reflection retarded in phase by π with respect to the first plane, thereby canceling the contribution from that plane. The cancellation of the (100) reflection occurs in the bcc lattice because the planes are identical in composition. A similar cancellation can easily be found in the hcp structure.

Structure Factor of the fcc Lattice

The basis of the fcc structure referred to the cubic cell has identical atoms at 000; $0\frac{11}{22}$; $\frac{1}{2}0\frac{1}{2}$; $\frac{11}{22}0$. Thus (46) becomes

$$\begin{aligned}S(v_1 v_2 v_3) &= f[1 + \exp[-i\pi(v_2 + v_3)] + \exp[-i\pi(v_1 + v_3)] \\ &\quad + \exp[-i\pi(v_1 + v_2)]] \quad .\end{aligned}\tag{48}$$

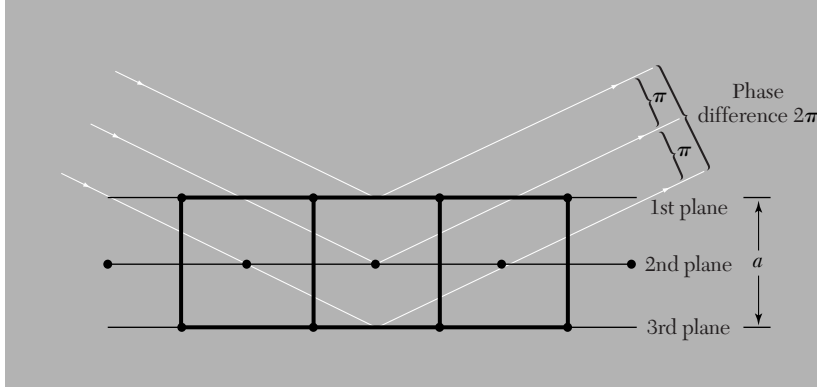


Figure 16 Explanation of the absence of a (100) reflection from a body-centered cubic lattice. The phase difference between successive planes is π , so that the reflected amplitude from two adjacent planes is $1 + e^{-i\pi} = 1 - 1 = 0$.

If all indices are even integers, $S = 4f$; similarly if all indices are odd integers. But if only one of the integers is even, two of the exponents will be odd multiples of $-i\pi$ and S will vanish. If only one of the integers is odd, the same argument applies and S will also vanish. Thus in the fcc lattice no reflections can occur for which the indices are partly even and partly odd.

The point is beautifully illustrated by Fig. 17: both KCl and KBr have an fcc lattice, but $n(\mathbf{r})$ for KCl simulates an sc lattice because the K^+ and Cl^- ions have equal numbers of electrons.

Atomic Form Factor

In the expression (46) for the structure factor, there occurs the quantity f_j , which is a measure of the scattering power of the j th atom in the unit cell. The value of f involves the number and distribution of atomic electrons, and the wavelength and angle of scattering of the radiation. We now give a classical calculation of the scattering factor.

The scattered radiation from a single atom takes account of interference effects within the atom. We defined the form factor in (42):

$$f_j = \int dV n_j(\mathbf{r}) \exp(-i\mathbf{G} \cdot \mathbf{r}) , \quad (49)$$

with the integral extended over the electron concentration associated with a single atom. Let $\mathbf{r}$ make an angle α with $\mathbf{G}$; then $\mathbf{G} \cdot \mathbf{r} = Gr \cos \alpha$. If the electron distribution is spherically symmetric about the origin, then

$$\begin{aligned} f_j &\equiv 2\pi \int dr r^2 d(\cos \alpha) n_j(r) \exp(-iGr \cos \alpha) \\ &= 2\pi \int dr r^2 n_j(r) \cdot \frac{e^{iGr} - e^{-iGr}}{iGr} , \end{aligned}$$

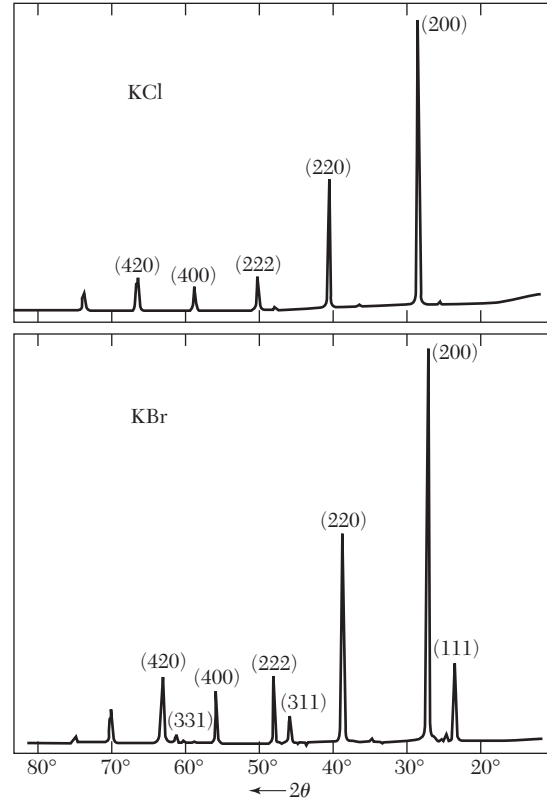


Figure 17 Comparison of x-ray reflections from KCl and KBr powders. In KCl the numbers of electrons of K^+ and Cl^- ions are equal. The scattering amplitudes $f(\text{K}^+)$ and $f(\text{Cl}^-)$ are almost exactly equal, so that the crystal looks to x-rays as if it were a monatomic simple cubic lattice of lattice constant $a/2$. Only even integers occur in the reflection indices when these are based on a cubic lattice of lattice constant a . In KBr the form factor of Br^- is quite different to that of K^+ , and all reflections of the fcc lattice are present. (Courtesy of R. van Nordstrand.)

after integration over $d(\cos \alpha)$ between -1 and 1 . Thus the form factor is given by

$$f_j = 4\pi \int dr n_j(r) r^2 = \frac{\sin Gr}{Gr} . \quad (50)$$

If the same total electron density were concentrated at $r = 0$, only $Gr = 0$ would contribute to the integrand. In this limit $(\sin Gr)/Gr = 1$, and

$$f_j = 4\pi \int dr n_j(r) r^2 = Z , \quad (51)$$

the number of atomic electrons. Therefore f is the ratio of the radiation amplitude scattered by the actual electron distribution in an atom to that scattered by one electron localized at a point. In the forward direction $G = 0$, and f reduces again to the value Z .

The overall electron distribution in a solid as seen in x-ray diffraction is fairly close to that of the appropriate free atoms. This statement does not mean that the outermost or valence electrons are not redistributed somewhat in forming the solid; it means only that the x-ray reflection intensities are represented well by the free atom values of the form factors and are not very sensitive to small redistributions of the electrons.

SUMMARY

- Various statements of the Bragg condition:

$$2d \sin \theta = n\lambda \quad ; \quad \Delta \mathbf{k} = \mathbf{G} \quad ; \quad 2\mathbf{k} \cdot \mathbf{G} = G^2 \quad .$$

- Laue conditions:

$$\mathbf{a}_1 \cdot \Delta \mathbf{k} = 2\pi v_1 \quad ; \quad \mathbf{a}_2 \cdot \Delta \mathbf{k} = 2\pi v_2 \quad ; \quad \mathbf{a}_3 \cdot \Delta \mathbf{k} = 2\pi v_3 \quad .$$

- The primitive translation vectors of the reciprocal lattice are

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} \quad ; \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} \quad ; \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} \quad .$$

Here $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are the primitive translation vectors of the crystal lattice.

- A reciprocal lattice vector has the form

$$\mathbf{G} = v_1 \mathbf{b}_1 + v_2 \mathbf{b}_2 + v_3 \mathbf{b}_3 \quad ,$$

where v_1, v_2, v_3 are integers or zero.

- The scattered amplitude in the direction $\mathbf{k}' = \mathbf{k} + \Delta \mathbf{k} = \mathbf{k} + \mathbf{G}$ is proportional to the geometrical structure factor:

$$S_{\mathbf{G}} \equiv \sum_j f_j \exp(-i\mathbf{r}_j \cdot \mathbf{G}) = \sum_j f_j \exp[-i2\pi(x_j v_1 + y_j v_2 + z_j v_3)] \quad ,$$

where j runs over the s atoms of the basis, and f_j is the atomic form factor (49) of the j th atom of the basis. The expression on the right-hand side is written for a reflection $(v_1 v_2 v_3)$, for which $\mathbf{G} = v_1 \mathbf{b}_1 + v_2 \mathbf{b}_2 + v_3 \mathbf{b}_3$.

- Any function invariant under a lattice translation $\mathbf{T}$ may be expanded in a Fourier series of the form

$$n(\mathbf{r}) = \sum_{\mathbf{G}} n_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r}) \quad .$$

- The first Brillouin zone is the Wigner-Seitz primitive cell of the reciprocal lattice. Only waves whose wavevector $\mathbf{k}$ drawn from the origin terminates on a surface of the Brillouin zone can be diffracted by the crystal.
- | | |
|------------------------|--------------------------------|
| <i>Crystal lattice</i> | <i>First Brillouin zone</i> |
| Simple cubic | Cube |
| Body-centered cubic | Rhombic dodecahedron (Fig. 13) |
| Face-centered cubic | Truncated octahedron (Fig. 15) |

Problems

1. **Interplanar separation.** Consider a plane hkl in a crystal lattice. (a) Prove that the reciprocal lattice vector $\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ is perpendicular to this plane. (b) Prove that the distance between two adjacent parallel planes of the lattice is $d(hkl) = 2\pi/|\mathbf{G}|$. (c) Show for a simple cubic lattice that $d^2 = a^2/(h^2 + k^2 + l^2)$.

2. **Hexagonal space lattice.** The primitive translation vectors of the hexagonal space lattice may be taken as

$$\mathbf{a}_1 = (3^{1/2}a/2)\hat{\mathbf{x}} + (a/2)\hat{\mathbf{y}} ; \quad \mathbf{a}_2 = -(3^{1/2}a/2)\hat{\mathbf{x}} + (a/2)\hat{\mathbf{y}} ; \quad \mathbf{a}_3 = c\hat{\mathbf{z}} .$$

- (a) Show that the volume of the primitive cell is $(3^{1/2}/2)a^2c$.
 (b) Show that the primitive translations of the reciprocal lattice are

$$\mathbf{b}_1 = (2\pi/3^{1/2}a)\hat{\mathbf{x}} + (2\pi/a)\hat{\mathbf{y}} ; \quad \mathbf{b}_2 = -(2\pi/3^{1/2}a)\hat{\mathbf{x}} + (2\pi/a)\hat{\mathbf{y}} ; \quad \mathbf{b}_3 = (2\pi/c)\hat{\mathbf{z}} ,$$

so that the lattice is its own reciprocal, but with a rotation of axes.

- (c) Describe and sketch the first Brillouin zone of the hexagonal space lattice.

3. **Volume of Brillouin zone.** Show that the volume of the first Brillouin zone is $(2\pi)^3/V_c$, where V_c is the volume of a crystal primitive cell. Hint: The volume of a Brillouin zone is equal to the volume of the primitive parallelepiped in Fourier space. Recall the vector identity $(\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{c} \cdot \mathbf{a} \times \mathbf{b})\mathbf{a}$.

4. **Width of diffraction maximum.** We suppose that in a linear crystal there are identical point scattering centers at every lattice point $\mathbf{p}_m = m\mathbf{a}$, where m is an integer. By analogy with (20), the total scattered radiation amplitude will be proportional to $F = \sum \exp[-im\mathbf{a} \cdot \Delta\mathbf{k}]$. The sum over M lattice points is

$$F = \frac{1 - \exp[-iM(\mathbf{a} \cdot \Delta\mathbf{k})]}{1 - \exp[-i(\mathbf{a} \cdot \Delta\mathbf{k})]} ,$$

by the use of the series

$$\sum_{m=0}^{M-1} x^m = \frac{1 - x^M}{1 - x} .$$

- (a) The scattered intensity is proportional to $|F|^2$. Show that

$$|F|^2 \equiv F^*F = \frac{\sin^2 \frac{1}{2} M(\mathbf{a} \cdot \Delta\mathbf{k})}{\sin^2 \frac{1}{2} (\mathbf{a} \cdot \Delta\mathbf{k})} .$$

- (b) We know that a diffraction maximum appears when $\mathbf{a} \cdot \Delta\mathbf{k} = 2\pi h$, where h is an integer. We change $\Delta\mathbf{k}$ slightly and define ϵ in $\mathbf{a} \cdot \Delta\mathbf{k} = 2\pi h + \epsilon$ such that ϵ gives the position of the first zero in $\sin \frac{1}{2} M(\mathbf{a} \cdot \Delta\mathbf{k})$. Show that $\epsilon = 2\pi/M$, so that the width of the diffraction maximum is proportional to $1/M$ and can be extremely narrow for macroscopic values of M . The same result holds true for a three-dimensional crystal.

5. **Structure factor of diamond.** The crystal structure of diamond is described in Chapter 1. The basis consists of eight atoms if the cell is taken as the conventional cube. (a) Find the structure factor S of this basis. (b) Find the zeros of S and show that the allowed reflections of the diamond structure satisfy $v_1 + v_2 + v_3 = 4n$, where all indices are even and n is any integer, or else all indices are odd (Fig. 18). (Notice that h, k, l may be written for v_1, v_2, v_3 and this is often done.)

6. **Form factor of atomic hydrogen.** For the hydrogen atom in its ground state, the number density is $n(r) = (\pi a_0^3)^{-1} \exp(-2r/a_0)$, where a_0 is the Bohr radius. Show that the form factor is $f_G = 16/(4 + G^2 a_0^2)^2$.

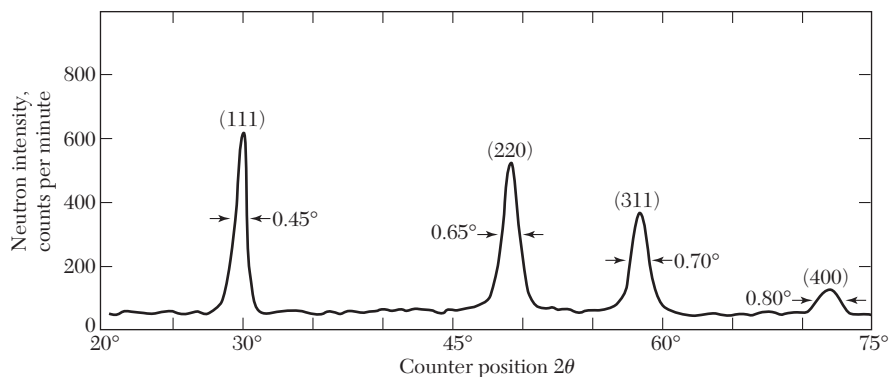


Figure 18 Neutron diffraction pattern for powdered diamond. (After G. Bacon.)

7. Diatomic line. Consider a line of atoms $ABAB \dots AB$, with an $A-B$ bond length of $\frac{1}{2}a$. The form factors are f_A, f_B for atoms A, B , respectively. The incident beam of x-rays is perpendicular to the line of atoms. (a) Show that the interference condition is $n\lambda = a \cos \theta$, where θ is the angle between the diffracted beam and the line of atoms. (b) Show that the intensity of the diffracted beam is proportional to $|f_A - f_B|^2$ for n odd, and to $|f_A + f_B|^2$ for n even. (c) Explain what happens if $f_A = f_B$.

3

Crystal Binding and Elastic Constants

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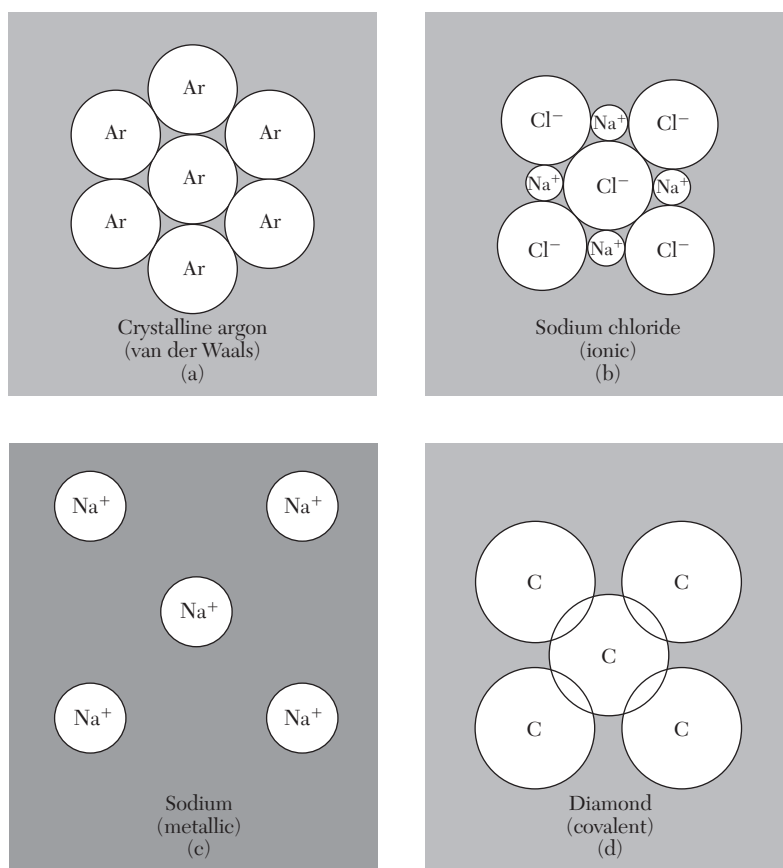


Figure 1 The principal types of crystalline binding. In (a) neutral atoms with closed electron shells are bound together weakly by the van der Waals forces associated with fluctuations in the charge distributions. In (b) electrons are transferred from the alkali atoms to the halogen atoms, and the resulting ions are held together by attractive electrostatic forces between the positive and negative ions. In (c) the valence electrons are taken away from each alkali atom to form a communal electron sea in which the positive ions are dispersed. In (d) the neutral atoms are bound together by the overlapping parts of their electron distributions.

In this chapter we are concerned with the question: What holds a crystal together? The attractive electrostatic interaction between the negative charges of the electrons and the positive charges of the nuclei is entirely responsible for the cohesion of solids. Magnetic forces have only a weak effect on cohesion, and gravitational forces are negligible. Specialized terms categorize distinctive situations: exchange energy, van der Waals forces, and covalent bonds. The observed differences between the forms of condensed matter are caused in the final analysis by differences in the distribution of the outermost electrons and the ion cores (Fig. 1).

The cohesive energy of a crystal is defined as the energy that must be added to the crystal to separate its components into neutral free atoms at rest, at infinite separation, with the same electronic configuration. The term lattice energy is used in the discussion of ionic crystals and is defined as the energy that must be added to the crystal to separate its component ions into free ions at rest at infinite separation.

Values of the cohesive energy of the crystalline elements are given in Table 1. Notice the wide variation in cohesive energy between different columns of the periodic table. The inert gas crystals are weakly bound, with cohesive energies less than a few percent of the cohesive energies of the elements in the C, Si, Ge . . . column. The alkali metal crystals have intermediate values of the cohesive energy. The transition element metals (in the middle columns) are quite strongly bound. The melting temperatures (Table 2) and bulk moduli (Table 3) vary roughly as the cohesive energies.

CRYSTALS OF INERT GASES

The inert gases form the simplest crystals. The electron distribution is very close to that of the free atoms. Their properties at absolute zero are summarized in Table 4. The crystals are transparent insulators, weakly bound, with low melting temperatures. The atoms have very high ionization energies (see Table 5). The outermost electron shells of the atoms are completely filled, and the distribution of electron charge in the free atom is spherically symmetric. In the crystal the inert gas atoms pack together as closely as possible¹: the

¹Zero-point motion of the atoms (kinetic energy at absolute zero) is a quantum effect that plays a dominant role in He³ and He⁴. They do not solidify at zero pressure even at absolute zero temperature. The average fluctuation at 0 K of a He atom from its equilibrium position is of the order of 30 to 40 percent of the nearest-neighbor distance. The heavier the atom, the less important the zero-point effects. If we omit zero-point motion, we calculate a molar volume of 9 cm³ mol⁻¹ for solid helium, as compared with the observed values of 27.5 and 36.8 cm³ mol⁻¹ for liquid He⁴ and liquid He³, respectively.

Table 1 Cohesive energies

Table 1 Cohesive energies												B	C	N	O	F	Ne
Li	Be	Energy required to form separated neutral atoms in their ground electronic state from the solid at 0 K at 1 atm. The data were supplied by Prof. Leo Brewer.										561	711.	474.	251.	81.0	1.92
158.	320.											5.81	7.37	4.92	2.60	0.84	0.020
1.63	3.32											134	170.	113.4	60.03	19.37	0.46
37.7	76.5																
Na	Mg											Al	Si	P	S	Cl	Ar
107.	145.	←————— kJ/mol —————→										327.	446.	331.	275.	135.	7.74
1.113	1.51	←————— eV/atom —————→										3.39	4.63	3.43	2.85	1.40	0.080
25.67	34.7	←————— kcal/mol —————→										78.1	106.7	79.16	65.75	32.2	1.85
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
90.1	178.	376	468.	512.	395.	282.	413.	424.	428.	336.	130	271.	372.	285.3	237	118.	11.2
0.934	1.84	3.90	4.85	5.31	4.10	2.92	4.28	4.39	4.44	3.49	1.35	2.81	3.85	2.96	2.46	1.22	0.116
21.54	42.5	89.9	111.8	122.4	94.5	67.4	98.7	101.3	102.4	80.4	31.04	64.8	88.8	68.2	56.7	28.18	2.68
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
82.2	166.	422.	603.	730.	658	661.	650.	554.	376.	284.	112.	243.	303.	265.	211	107.	15.9
0.852	1.72	4.37	6.25	7.57	6.82	6.85	6.74	5.75	3.89	2.95	1.16	2.52	3.14	2.75	2.19	1.11	0.16
19.64	39.7	100.8	144.2	174.5	157.2	158.	155.4	132.5	89.8	68.0	26.73	58.1	72.4	63.4	50.34	25.62	3.80
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
77.6	183.	431.	621.	782.	859.	775.	788.	670.	564.	368.	65.	182.	196.	210.	144.		19.5
0.804	1.90	4.47	6.44	8.10	8.90	8.03	8.17	6.94	5.84	3.81	0.67	1.88	2.03	2.18	1.50		0.202
18.54	43.7	103.1	148.4	186.9	205.2	185.2	188.4	160.1	134.7	87.96	15.5	43.4	46.78	50.2	34.5		4.66
Fr	Ra	Ac															
	160.	410.	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
	1.66	4.25	417.	357.	328.		206.	179.	400.	391.	294.	302.	317.	233.	154.	428.	
	38.2	98.	4.32	3.70	3.40		2.14	1.86	4.14	4.05	3.04	3.14	3.29	2.42	1.60	4.43	
			99.7	85.3	78.5		49.3	42.8	95.5	93.4	70.2	72.3	75.8	55.8	37.1	102.2	
			Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	
			598.		536.	456	347.	264.	385								
			6.20		5.55	4.73	3.60	2.73	3.99								
			142.9		128.	109.	83.0	63.	92.1								

Table 2 Melting points, in K.
(After R. H. Lamoreaux)

Li 453.7	Be 1562											B 2365	C	N 63.15	O 54.36	F 53.48	Ne 24.56
Na 371.0	Mg 922											Al 933.5	Si 1687	P w 317 r 863	S 388.4	Cl 172.2	Ar 83.81
K 336.3	Ca 1113	Sc 1814	Ti 1946	V 2202	Cr 2133	Mn 1520	Fe 1811	Co 1770	Ni 1728	Cu 1358	Zn 692.7	Ga 302.9	Ge 1211	As 1089	Se 494	Br 265.9	Kr 115.8
Rb 312.6	Sr 1042	Y 1801	Zr 2128	Nb 2750	Mo 2895	Tc 2477	Ru 2527	Rh 2236	Pd 1827	Ag 1235	Cd 594.3	In 429.8	Sn 505.1	Sb 903.9	Te 722.7	I 386.7	Xe 161.4
Cs 301.6	Ba 1002	La 1194	Hf 2504	Ta 3293	W 3695	Re 3459	Os 3306	Ir 2720	Pt 2045	Au 1338	Hg 234.3	Tl 577	Pb 600.7	Bi 544.6	Po 527	At	Rn
Fr	Ra 973	Ac 1324															
			Ce 1072	Pr 1205	Nd 1290	Pm	Sm 1346	Eu 1091	Gd 1587	Tb 1632	Dy 1684	Ho 1745	Er 1797	Tm 1820	Yb 1098	Lu 1938	
			Th 2031	Pa 1848	U 1406	Np 910	Pu 913	Am 1449	Cm 1613	Bk 1562	Cf	Es	Fm	Md	No	Lw	

Table 4 Properties of inert gas crystals

(Extrapolated to 0 K and zero pressure)

	Nearest-neighbor distance, in Å	Experimental cohesive energy		Melting point, K	Ionization potential of free atom, eV	Parameters in Lennard-Jones potential, Eq. 10	
		kJ/mol	eV/atom			ϵ , in 10^{-16} erg	σ , in Å
He	(liquid at zero pressure)				24.58	14	2.56
Ne	3.13	1.88	0.02	24.56	21.56	50	2.74
Ar	3.76	7.74	0.080	83.81	15.76	167	3.40
Kr	4.01	11.2	0.116	115.8	14.00	225	3.65
Xe	4.35	16.0	0.17	161.4	12.13	320	3.98

crystal structures (Fig. 2) are all cubic close-packed (fcc), except He³ and He⁴.

What holds an inert gas crystal together? The electron distribution in the crystal is not significantly distorted from the electron distribution around the free atoms because not much energy is available to distort the free atom charge distributions. The cohesive energy of an atom in the crystal is only 1 percent or less of the ionization energy of an atomic electron. Part of this distortion gives the van der Waals interaction.

Van der Waals-London Interaction

Consider two identical inert gas atoms at a separation R large in comparison with the radii of the atoms. What interactions exist between the two neutral atoms? If the charge distributions on the atoms were rigid, the interaction between atoms would be zero, because the electrostatic potential of a spherical distribution of electronic charge is canceled outside a neutral atom by the electrostatic potential of the charge on the nucleus. Then the inert gas atoms could show no cohesion and could not condense. But the atoms induce dipole moments in each other, and the induced moments cause an attractive interaction between the atoms.

As a model, we consider two identical linear harmonic oscillators 1 and 2 separated by R . Each oscillator bears charges $\pm e$ with separations x_1 and x_2 , as in Fig. 3. The particles oscillate along the x axis. Let p_1 and p_2 denote the momenta. The force constant is C . Then the hamiltonian of the unperturbed system is

$$\mathcal{H}_0 = \frac{1}{2m} p_1^2 + \frac{1}{2} C x_1^2 + \frac{1}{2m} p_2^2 + \frac{1}{2} C x_2^2 . \quad (1)$$

Each uncoupled oscillator is assumed to have the frequency ω_0 of the strongest optical absorption line of the atom. Thus $C = m\omega_0^2$.

Table 5 Ionization energies

The total energy required to remove the first two electrons is the sum of the first and second ionization potentials. (Source: National Bureau of Standards Circular 467.)

Table 5 Ionization energies

The total energy required to remove the first two electrons is the sum of the first and second ionization potentials. (Source: National Bureau of Standards Circular 467.)

H	13.595																	He	24.58 78.98																
Li	5.39 81.01	Be	9.32 27.53															Ne	21.56 62.63																
Na	5.14 52.43	Mg	7.64 22.67															Ar	15.76 43.38																
				← ————— Energy to remove one electron, in eV ————— →																															
				← ————— Energy to remove two electrons, in eV ————— →																															
K	4.34 36.15	Ca	6.11 17.98	Sc	6.56 19.45	Ti	6.83 20.46	V	6.74 21.39	Cr	6.76 23.25	Mn	7.43 23.07	Fe	7.90 24.08	Co	7.86 24.91	Ni	7.63 25.78	Cu	7.72 27.93	Zn	9.39 27.35	Ga	6.00 26.51	Ge	7.88 23.81	As	9.81 30.0	Se	9.75 31.2	Br	11.84 33.4	Kr	14.00 38.56
Rb	4.18 31.7	Sr	5.69 16.72	Y	6.5 18.9	Zr	6.95 20.98	Nb	6.77 21.22	Mo	7.18 23.25	Tc	7.28 22.54	Ru	7.36 24.12	Rh	7.46 25.53	Pd	8.33 27.75	Ag	7.57 29.05	Cd	8.99 25.89	In	5.78 24.64	Sn	7.34 21.97	Sb	8.64 25.1	Te	9.01 27.6	I	10.45 29.54	Xe	12.13 33.3
Cs	3.89 29.0	Ba	5.21 15.21	La	5.61 17.04	Hf	7. 22.	Ta	7.88 24.1	W	7.98 25.7	Re	7.87 24.5	Os	8.7 26.	Ir	9. 27.52	Pt	8.96 29.7	Au	9.22 29.18	Hg	10.43 29.18	Tl	6.11 26.53	Pb	7.41 22.44	Bi	7.29 23.97	Po	8.43	At		Rn	10.74
Fr		Ra	5.28 15.42	Ac	6.9 19.0																														
				Ce	6.91	Pr	5.76	Nd	6.31	Pm		Sm	5.6	Eu	5.67	Gd	6.16	Tb	6.74	Dy	6.82	Ho		Er		Tm		Yb	6.2	Lu	5.0				
				Th		Pa		U	4.	Np		Pu		Am		Cm		Bk		Cf		Es		Fm		Md		No		Lr					

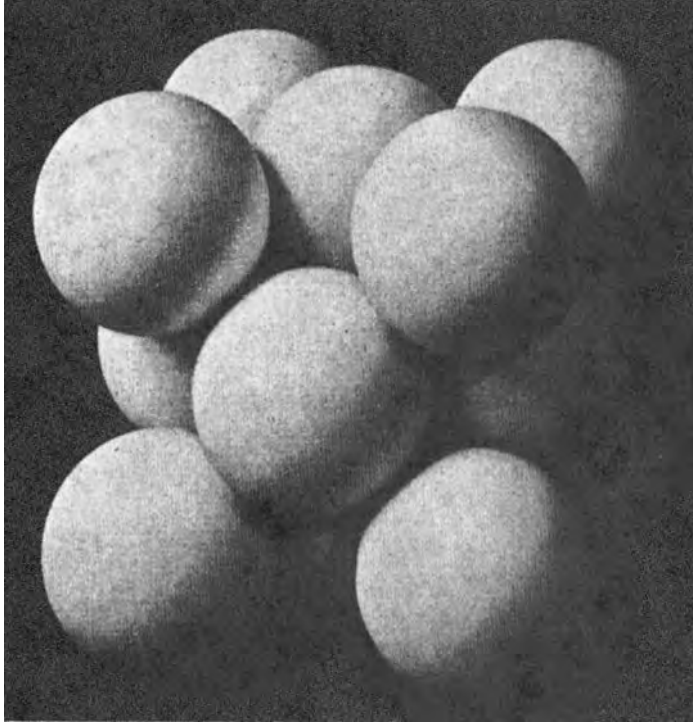


Figure 2 Cubic close-packed (fcc) crystal structure of the inert gases Ne, Ar, Kr, and Xe. The lattice parameters of the cubic cells are 4.46, 5.31, 5.64, and 6.13 Å, respectively, at 4 K.



Figure 3 Coordinates of the two oscillators.

Let $\mathcal{H}_1$ be the coulomb interaction energy of the two oscillators. The geometry is shown in the figure. The internuclear coordinate is R . Then

$$(\text{CGS}) \quad \mathcal{H}_1 = \frac{e^2}{R} + \frac{e^2}{R + x_1 - x_2} - \frac{e^2}{R + x_1} - \frac{e^2}{R - x_2} ; \quad (2)$$

in the approximation $|x_1|, |x_2| \ll R$ we expand (2) to obtain in lowest order:

$$\mathcal{H}_1 \cong -\frac{2e^2 x_1 x_2}{R^3} . \quad (3)$$

The total hamiltonian with the approximate form (3) for $\mathcal{H}_1$ can be diagonalized by the normal mode transformation

$$x_s \equiv \frac{1}{\sqrt{2}}(x_1 + x_2) ; \quad x_a \equiv \frac{1}{\sqrt{2}}(x_1 - x_2) , \quad (4)$$

or, on solving for x_1 and x_2 ,

$$x_1 = \frac{1}{\sqrt{2}}(x_s + x_a) ; \quad x_2 = \frac{1}{\sqrt{2}}(x_s - x_a) . \quad (5)$$

The subscripts s and a denote symmetric and antisymmetric modes of motion. Further, we have the momenta p_s, p_a associated with the two modes:

$$p_1 \equiv \frac{1}{\sqrt{2}}(p_s + p_a) ; \quad p_2 \equiv \frac{1}{\sqrt{2}}(p_s - p_a) . \quad (6)$$

The total hamiltonian $\mathcal{H}_0 + \mathcal{H}_1$ after the transformations (5) and (6) is

$$\mathcal{H} = \left[\frac{1}{2m}p_s^2 + \frac{1}{2}\left(C - \frac{2e^2}{R^3}\right)x_s^2 \right] + \left[\frac{1}{2m}p_a^2 + \frac{1}{2}\left(C + \frac{2e^2}{R^3}\right)x_a^2 \right] . \quad (7)$$

The two frequencies of the coupled oscillators are found by inspection of (7) to be

$$\omega = \left[\left(C \pm \frac{2e^2}{R^3} \right) / m \right]^{1/2} = \omega_0 \left[1 \pm \frac{1}{2} \left(\frac{2e^2}{CR^3} \right) - \frac{1}{8} \left(\frac{2e^2}{CR^3} \right)^2 + \dots \right] , \quad (8)$$

with ω_0 given by $(C/m)^{1/2}$. In (8) we have expanded the square root.

The zero point energy of the system is $\frac{1}{2}\hbar(\omega_s + \omega_a)$; because of the interaction the sum is lowered from the uncoupled value $2 \cdot \frac{1}{2}\hbar\omega_0$ by

$$\Delta U = \frac{1}{2}\hbar(\Delta\omega_s + \Delta\omega_a) = -\hbar\omega_0 \cdot \frac{1}{8} \left(\frac{2e^2}{CR^3} \right)^2 = -\frac{A}{R^6} . \quad (9)$$

This attractive interaction varies as the minus sixth power of the separation of the two oscillators.

This is called the van der Waals interaction, known also as the London interaction or the induced dipole-dipole interaction. It is the principal attractive interaction in crystals of inert gases and also in crystals of many organic molecules. The interaction is a quantum effect, in the sense that $\Delta U \rightarrow 0$ as $\hbar \rightarrow 0$. Thus the zero point energy of the system is lowered by the dipole-dipole coupling of Eq. (3). The van der Waals interaction does not depend for its existence on any overlap of the charge densities of the two atoms.

An approximate value of the constant A in (9) for identical atoms is given by $\hbar\omega_0\alpha^2$, where $\hbar\omega_0$ is the energy of the strongest optical absorption line and α is the electronic polarizability (Chapter 15).

Repulsive Interaction

As the two atoms are brought together, their charge distributions gradually overlap (Fig. 4), thereby changing the electrostatic energy of the system. At sufficiently close separations the overlap energy is repulsive, in large part because of the **Pauli exclusion principle**. The elementary statement of the principle is that two electrons cannot have all their quantum numbers equal. When the charge distributions of two atoms overlap, there is a tendency for electrons from atom B to occupy in part states of atom A already occupied by electrons of atom A , and vice versa.

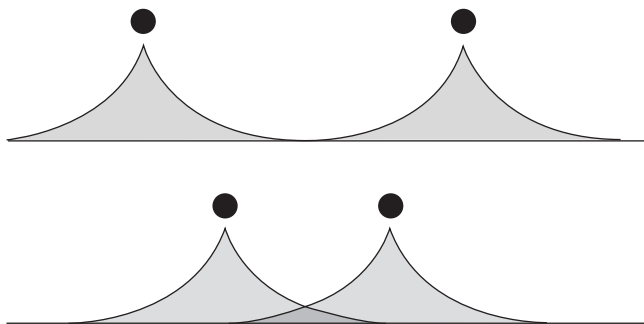


Figure 4 Electronic charge distributions overlap as atoms approach. The solid circles denote the nuclei.

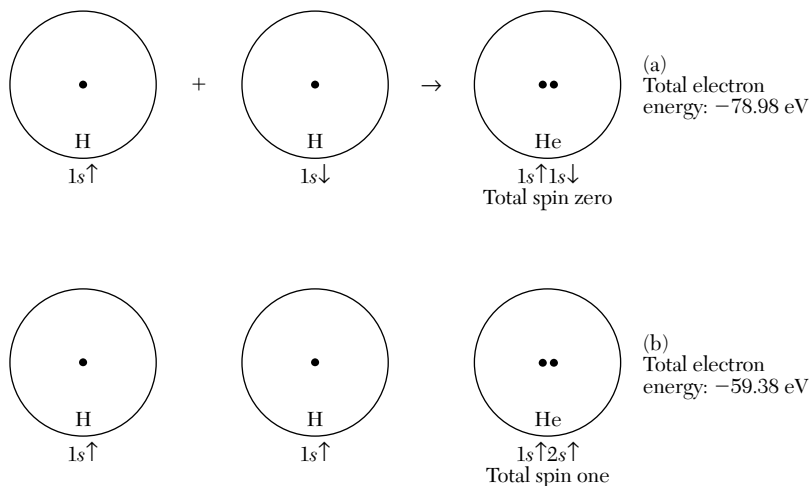


Figure 5 The effect of Pauli principle on the repulsive energy: in an extreme example, two hydrogen atoms are pushed together until the protons are almost in contact. The energy of the electron system alone can be taken from observations on atomic He, which has two electrons. In (a) the electrons have antiparallel spins and the Pauli principle has no effect: the electrons are bound by -78.98 eV . In (b) the spins are parallel: the Pauli principle forces the promotion of an electron from a $1s \uparrow$ orbital of H to a $2s \uparrow$ orbital of He. The electrons now are bound by -59.38 eV , less than (a) by 19.60 eV . This is the amount by which the Pauli principle has increased the repulsion. We have omitted the repulsive coulomb energy of the two protons, which is the same in both (a) and (b).

The Pauli principle prevents multiple occupancy, and electron distributions of atoms with closed shells can overlap only if accompanied by the partial promotion of electrons to unoccupied high energy states of the atoms. Thus the electron overlap increases the total energy of the system and gives a repulsive contribution to the interaction. An extreme example in which the overlap is complete is shown in Fig. 5.

We make no attempt here to evaluate the repulsive interaction² from first principles. Experimental data on the inert gases can be fitted well by an empirical repulsive potential of the form B/R^{12} , where B is a positive constant, when used

²The overlap energy naturally depends on the radial distribution of charge about each atom. The mathematical calculation is always complicated even if the charge distribution is known.

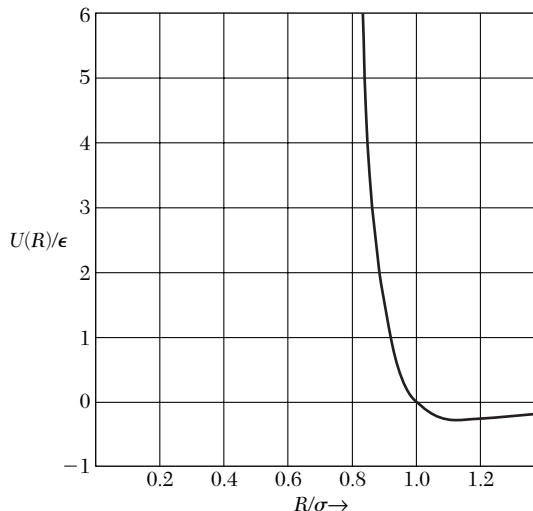


Figure 6 Form of the Lennard-Jones potential (10) which describes the interaction of two inert gas atoms. The minimum occurs at $R/\sigma = 2^{1/6} \cong 1.12$. Notice how steep the curve is inside the minimum, and how flat it is outside the minimum. The value of U at the minimum is $-\epsilon$; and $U = 0$ at $R = \sigma$.

together with a long-range attractive potential of the form of (9). The constants A and B are empirical parameters determined from independent measurements made in the gas phase; the data used include the virial coefficients and the viscosity. It is usual to write the total potential energy of two atoms at separation R as

$$U(R) = 4\epsilon \left[\left(\frac{\sigma}{R} \right)^{12} - \left(\frac{\sigma}{R} \right)^6 \right], \quad (10)$$

where ϵ and σ are the new parameters, with $4\epsilon\sigma^6 = A$ and $4\epsilon\sigma^{12} = B$. The potential (10) is known as the Lennard-Jones potential, Fig. 6. The force between the two atoms is given by $-dU/dR$. Values of ϵ and σ given in Table 4 can be obtained from gas-phase data, so that calculations on properties of the solid do not involve disposable parameters.

Other empirical forms for the repulsive interaction are widely used, in particular the exponential form $\lambda \exp(-R/\rho)$, where ρ is a measure of the range of the interaction. This is generally as easy to handle analytically as the inverse power law form.

Equilibrium Lattice Constants

If we neglect the kinetic energy of the inert gas atoms, the cohesive energy of an inert gas crystal is given by summing the Lennard-Jones potential (10) over all pairs of atoms in the crystal. If there are N atoms in the crystal, the total potential energy is

$$U_{\text{tot}} = \frac{1}{2}N(4\epsilon) \left[\sum_j' \left(\frac{\sigma}{p_{ij}R} \right)^{12} - \sum_j' \left(\frac{\sigma}{p_{ij}R} \right)^6 \right], \quad (11)$$

where $p_{ij}R$ is the distance between reference atom i and any other atom j , expressed in terms of the nearest-neighbor distance R . The factor $\frac{1}{2}$ occurs with the N to compensate for counting twice each pair of atoms.

The summations in (11) have been evaluated, and for the fcc structure

$$\sum_j' p_{ij}^{-12} = 12.13188 \ ; \quad \sum_j' p_{ij}^{-6} = 14.45392 \ . \quad (12)$$

There are 12 nearest-neighbor sites in the fcc structure; we see that the series are rapidly converging and have values not far from 12. The nearest neighbors contribute most of the interaction energy of inert gas crystals. The corresponding sums for the hcp structure are 12.13229 and 14.45489.

If we take U_{tot} in (11) as the total energy of the crystal, the equilibrium value R_0 is given by requiring that U_{tot} be a minimum with respect to variations in the nearest-neighbor distance R :

$$\frac{dU_{\text{tot}}}{dR} = 0 = -2N\epsilon \left[(12)(12.13) \frac{\sigma^{12}}{R^{13}} - (6)(14.45) \frac{\sigma^6}{R^7} \right] , \quad (13)$$

whence

$$R_0/\sigma = 1.09 \ , \quad (14)$$

the same for all elements with an fcc structure. The observed values of R_0/σ , using the independently determined values of σ given in Table 4, are:

	Ne	Ar	Kr	Xe
R_0/σ	1.14	1.11	1.10	1.09

The agreement with (14) is remarkable. The slight departure of R_0/σ for the lighter atoms from the universal value 1.09 predicted for inert gases can be explained by zero-point quantum effects. From measurements on the gas phase we have predicted the lattice constant of the crystal.

Cohesive Energy

The cohesive energy of inert gas crystals at absolute zero and at zero pressure is obtained by substituting (12) and (14) in (11):

$$U_{\text{tot}}(R) = 2N\epsilon \left[(12.13) \left(\frac{\sigma}{R} \right)^{12} - (14.45) \left(\frac{\sigma}{R} \right)^6 \right] , \quad (15)$$

and, at $R = R_0$,

$$U_{\text{tot}}(R_0) = -(2.15)(4N\epsilon) \ , \quad (16)$$

the same for all inert gases. This is the calculated cohesive energy when the atoms are at rest. Quantum-mechanical corrections act to reduce the binding by 28, 10, 6, and 4 percent of Eq. (16) for Ne, Ar, Kr, and Xe, respectively.

The heavier the atom, the smaller the quantum correction. We can understand the origin of the quantum correction by consideration of a simple model in which an atom is confined by fixed boundaries. If the particle has the quantum wavelength λ , where λ is determined by the boundaries, then the particle has kinetic energy $p^2/2M = (h/\lambda)^2/2M$ with the de Broglie relation $p = h/\lambda$ for the connection between the momentum and the wavelength of a particle. On this model the quantum zero-point correction to the energy is inversely proportional to the mass. The final calculated cohesive energies agree with the experimental values of Table 4 within 1 to 7 percent.

One consequence of the quantum kinetic energy is that a crystal of the isotope Ne^{20} is observed to have a larger lattice constant than a crystal of Ne^{22} . The higher quantum kinetic energy of the lighter isotope expands the lattice because the kinetic energy is reduced by expansion. The observed lattice constants (extrapolated to absolute zero from 2.5 K) are Ne^{20} , 4.4644 Å; Ne^{22} , 4.4559 Å.

IONIC CRYSTALS

Ionic crystals are made up of positive and negative ions. The ionic bond results from the electrostatic interaction of oppositely charged ions. Two common crystal structures found for ionic crystals, the sodium chloride and the cesium chloride structures, were shown in Chapter 1.

The electronic configurations of all ions of a simple ionic crystal correspond to closed electronic shells, as in the inert gas atoms. In lithium fluoride the configuration of the neutral atoms are, according to the periodic table in the front endpapers of this book, Li: $1s^2 2s$, F: $1s^2 2s^2 2p^5$. The singly charged ions have the configurations $\text{Li}^+ : 1s^2$, $\text{F}^- : 1s^2 2s^2 2p^6$, as for helium and neon, respectively. Inert gas atoms have closed shells, and the charge distributions are spherically symmetric. We expect that the charge distributions on each ion in an ionic crystal will have approximately spherical symmetry, with some distortion near the region of contact with neighboring atoms. This picture is confirmed by x-ray studies of electron distributions (Fig. 7).

A quick estimate suggests that we are not misguided in looking to electrostatic interactions for a large part of the binding energy of an ionic crystal. The distance between a positive ion and the nearest negative ion in crystalline sodium chloride is 2.81×10^{-8} cm, and the attractive coulomb part of the potential energy of the two ions by themselves is 5.1 eV. This value may be compared (Fig. 8) with the experimental value of 7.9 eV per molecular unit for the lattice energy of crystalline NaCl with respect to separated Na^+ and Cl^- ions. We now calculate the energy more closely.

Electrostatic or Madelung Energy

The long-range interaction between ions with charge $\pm q$ is the electrostatic interaction $\pm q^2/r$, attractive between ions of opposite charge and repulsive

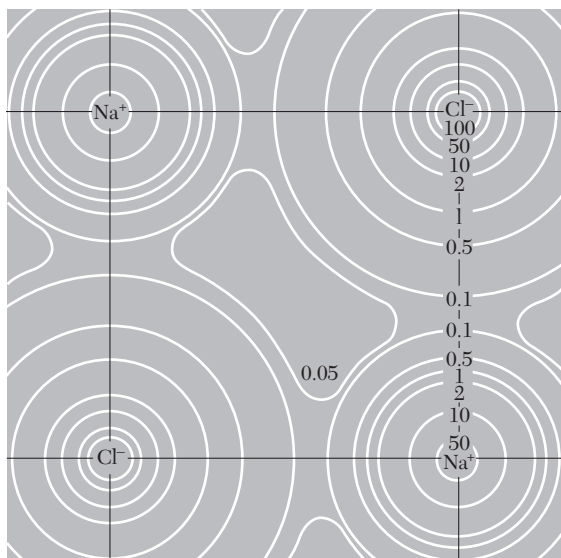


Figure 7 Electron density distribution in the base plane of NaCl, after x-ray studies by G. Schoknecht. The numbers on the contours give the relative electron concentration.

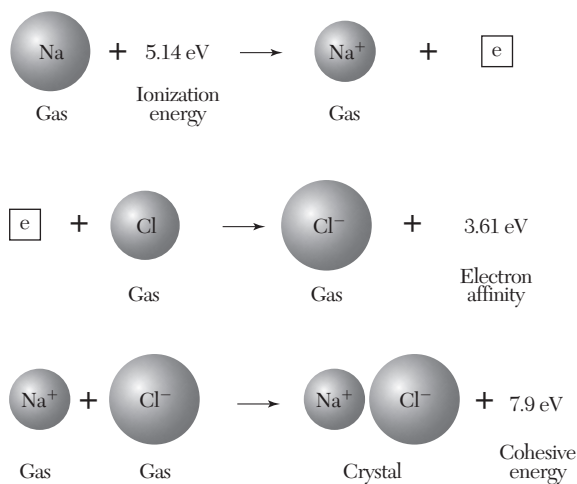


Figure 8 The energy per molecule unit of a crystal of sodium chloride is $(7.9 - 5.1 + 3.6) = 6.4$ eV lower than the energy of separated neutral atoms. The lattice energy with respect to separated ions is 7.9 eV per molecule unit. All values on the figure are experimental. Values of the ionization energy are given in Table 5, and values of the electron affinity are given in Table 6.

between ions of the same charge. The ions arrange themselves in whatever crystal structure gives the strongest attractive interaction compatible with the repulsive interaction at short distances between ion cores. The repulsive interactions between ions with inert gas configurations are similar to those between inert gas atoms. The van der Waals part of the attractive interaction in ionic crystals makes a relatively small contribution to the cohesive energy in ionic crystals, of the order of 1 or 2 percent. The main contribution to the binding energy of ionic crystals is electrostatic and is called the **Madelung energy**.

Table 6 Electron affinities of negative ions

The electron affinity is positive for a stable negative ion.

Atom	Electron affinity energy eV	Atom	Electron affinity energy eV
H	0.7542	Si	1.39
Li	0.62	P	0.74
C	1.27	S	2.08
O	1.46	Cl	3.61
F	3.40	Br	3.36
Na	0.55	I	3.06
Al	0.46	K	0.50

Source: H. Hotop and W. C. Lineberger, J. Phys. Chem. Ref. Data 4, 539 (1975).

If U_{ij} is the interaction energy between ions i and j , we define a sum U_i which includes all interactions involving the ion i :

$$U_i = \sum_j' U_{ij} , \quad (17)$$

where the summation includes all ions except $j = i$. We suppose that U_{ij} may be written as the sum of a central field repulsive potential of the form $\lambda \exp(-r/\rho)$, where λ and ρ are empirical parameters, and a coulomb potential $\pm q^2/r$. Thus

$$(CGS) \quad U_{ij} = \lambda \exp(-r_{ij}/\rho) \pm q^2/r_{ij} , \quad (18)$$

where the $+$ sign is taken for the like charges and the $-$ sign for unlike charges. In SI units the coulomb interaction is $\pm q^2/4\pi\epsilon_0 r$; we write this section in CGS units in which the coulomb interaction is $\pm q^2/r$.

The repulsive term describes the fact that each ion resists overlap with the electron distributions of neighboring ions. We treat the strength λ and range ρ as constants to be determined from observed values of the lattice constant and compressibility; we have used the exponential form of the empirical repulsive potential rather than the R^{-12} form used for the inert gases. The change is made because it may give a better representation of the repulsive interaction. For the ions, we do not have gas-phase data available to permit the independent determination of λ and ρ . We note that ρ is a measure of the range of the repulsive interaction; when $r = \rho$, the repulsive interaction is reduced to e^{-1} of the value at $r = 0$.

In the NaCl structure the value of U_i does not depend on whether the reference ion i is a positive or a negative ion. The sum in (17) can be arranged to converge rapidly, so that its value will not depend on the site of the reference ion in the crystal, as long as it is not near the surface. We neglect surface effects

and write the total lattice energy U_{tot} of a crystal composed of N molecules or $2N$ ions as $U_{\text{tot}} = NU_i$. Here N , rather than $2N$, occurs because we must count each *pair* of interactions only once or each bond only once. The total lattice energy is defined as the energy required to separate the crystal into individual ions at an infinite distance apart.

It is convenient again to introduce quantities p_{ij} such that $r_{ij} \equiv p_{ij}R$, where R is the nearest-neighbor separation in the crystal. If we include the repulsive interaction only among nearest neighbors, we have

$$(CGS) \quad U_{ij} = \begin{cases} \lambda \exp(-R/p) - \frac{q^2}{R} & (\text{nearest neighbors}) \\ \pm \frac{1}{p_{ij}} \frac{q^2}{R} & (\text{otherwise}). \end{cases} \quad (19)$$

Thus

$$(CGS) \quad U_{\text{tot}} = NU_i = N \left(z\lambda e^{-R/p} - \frac{\alpha q^2}{R} \right), \quad (20)$$

where z is the number of nearest neighbors of any ion and

$$\boxed{\alpha \equiv \sum_j' \frac{(\pm)}{p_{ij}} \equiv \text{Madelung constant}} \quad (21)$$

The sum should include the nearest-neighbor contribution, which is just z . The $(\pm)$ sign is discussed just before (25). The value of the Madelung constant is of central importance in the theory of an ionic crystal. Methods for its calculation are discussed next.

At the equilibrium separation $dU_{\text{tot}}/dR = 0$, so that

$$(CGS) \quad N \frac{dU_i}{dR} = -\frac{Nz\lambda}{\rho} \exp(-R/\rho) + \frac{N\alpha q^2}{R^2} = 0, \quad (22)$$

or

$$(CGS) \quad R_0^2 \exp(-R_0/\rho) = \rho \alpha q^2 / z\lambda. \quad (23)$$

This determines the equilibrium separation R_0 if the parameters ρ , λ of the repulsive interaction are known. For SI, replace q^2 by $q^2/4\pi\epsilon_0$.

The total lattice energy of the crystal of $2N$ ions at their equilibrium separation R_0 may be written, using (20) and (23), as

$$(CGS) \quad U_{\text{tot}} = -\frac{N\alpha q^2}{R_0} \left(1 - \frac{\rho}{R_0} \right). \quad (24)$$

The term $-N\alpha q^2/R_0$ is the Madelung energy. We shall find that ρ is of the order of $0.1R_0$, so that the repulsive interaction has a very short range.

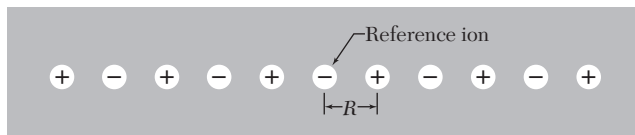


Figure 9 Line of ions of alternating signs, with distance R between ions.

Evaluation of the Madelung Constant

The first calculation of the coulomb energy constant α was made by Madelung. A powerful general method for lattice sum calculations was developed by Ewald and is developed in Appendix B. Computers are now used for the calculations.

The definition of the Madelung constant α is, by (21),

$$\alpha = \sum_j' \frac{(\pm)}{p_{ij}} .$$

For (20) to give a stable crystal it is necessary that α be positive. If we take the reference ion as a negative charge, the plus sign will apply to positive ions and the minus sign to negative ions.

An equivalent definition is

$$\frac{\alpha}{R} = \sum_j' \frac{(\pm)}{r_j} , \quad (25)$$

where r_j is the distance of the j th ion from the reference ion and R is the nearest-neighbor distance. The value given for α will depend on whether it is defined in terms of the nearest-neighbor distance R or in terms of the lattice parameter a or in terms of some other relevant length.

As an example, we compute the Madelung constant for the infinite line of ions of alternating sign in Fig. 9. Pick a negative ion as reference ion, and let R denote the distance between adjacent ions. Then

$$\frac{\alpha}{R} = 2 \left[\frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \cdots \right] ,$$

or

$$\alpha = 2 \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right] ;$$

the factor 2 occurs because there are two ions, one to the right and one to the left, at equal distances r_j . We sum this series by the expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots .$$

Thus the Madelung constant for the one-dimensional chain is $\alpha = 2 \ln 2$.

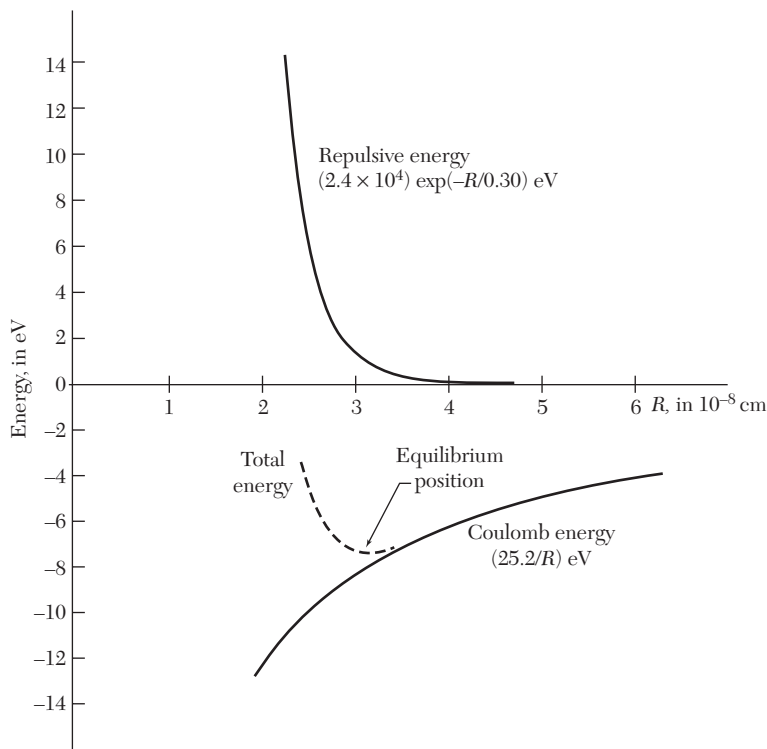


Figure 10 Energy per molecule of KCl crystal, showing Madelung (coulomb) and repulsive contributions.

In three dimensions the series presents greater difficulty. It is not possible to write down the successive terms by a casual inspection. More important, the series will not converge unless the successive terms in the series are arranged so that the contributions from the positive and negative terms nearly cancel.

Typical values of the Madelung constant are listed below, based on unit charges and *referred to the nearest-neighbor distance*:

Structure	α
Sodium chloride, NaCl	1.747565
Cesium chloride, CsCl	1.762675
Zinc blende, cubic ZnS	1.6381

The Madelung and repulsive contributions to the binding of a KCl crystal are shown in Fig. 10. Properties of alkali halide crystals having the sodium chloride structure are given in Table 7. The calculated values of the lattice energy are in exceedingly good agreement with the observed values.

Table 7 Properties of alkali halide crystals with the NaCl structure

All values (except those in square brackets) at room temperature and atmospheric pressure, with no correction for changes in R_0 and U from absolute zero. Values in square brackets at absolute zero temperature and zero pressure, from private communication by L. Brewer.

	Nearest-neighbor separation R_0 in Å	Bulk modulus B , in 10^{11} dyn/cm ² or 10^{10} N/m ²	Repulsive energy parameter $z\lambda$, in 10^{-8} erg	Repulsive range parameter ρ , in Å	Lattice energy compared to free ions, in kcal/mol	
					Experimental	Calculated
LiF	2.014	6.71	0.296	0.291	242.3[246.8]	242.2
LiCl	2.570	2.98	0.490	0.330	198.9[201.8]	192.9
LiBr	2.751	2.38	0.591	0.340	189.8	181.0
LiI	3.000	(1.71)	0.599	0.366	177.7	166.1
NaF	2.317	4.65	0.641	0.290	214.4[217.9]	215.2
NaCl	2.820	2.40	1.05	0.321	182.6[185.3]	178.6
NaBr	2.989	1.99	1.33	0.328	173.6[174.3]	169.2
NaI	3.237	1.51	1.58	0.345	163.2[162.3]	156.6
KF	2.674	3.05	1.31	0.298	189.8[194.5]	189.1
KCl	3.147	1.74	2.05	0.326	165.8[169.5]	161.6
KBr	3.298	1.48	2.30	0.336	158.5[159.3]	154.5
KI	3.533	1.17	2.85	0.348	149.9[151.1]	144.5
RbF	2.815	2.62	1.78	0.301	181.4	180.4
RbCl	3.291	1.56	3.19	0.323	159.3	155.4
RbBr	3.445	1.30	3.03	0.338	152.6	148.3
RbI	3.671	1.06	3.99	0.348	144.9	139.6

Data from various tables by M. P. Tosi, Solid State Physics **16**, 1 (1964).

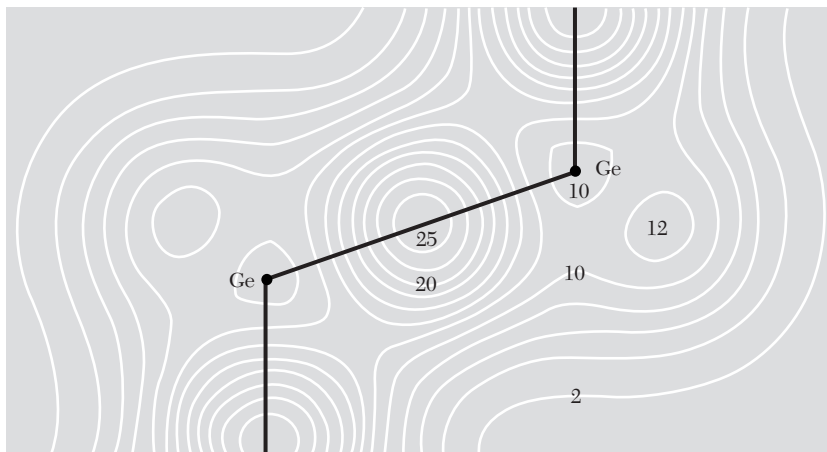


Figure 11 Calculated valence electron concentration in germanium. The numbers on the contours give the electron concentration per primitive cell, with four valence electrons per atom (eight electrons per primitive cell). Note the high concentration midway along the Ge-Ge bond, as we expect for covalent bonding. (After J. R. Chelikowsky and M. L. Cohen.)

COVALENT CRYSTALS

The covalent bond is the classical electron pair or homopolar bond of chemistry, particularly of organic chemistry. It is a strong bond: the bond between two carbon atoms in diamond with respect to separated neutral atoms is comparable with the bond strength in ionic crystals.

The covalent bond is usually formed from two electrons, one from each atom participating in the bond. The electrons forming the bond tend to be partly localized in the region between the two atoms joined by the bond. The spins of the two electrons in the bond are antiparallel.

The covalent bond has strong directional properties (Fig. 11). Thus carbon, silicon, and germanium have the diamond structure, with atoms joined to four nearest neighbors at tetrahedral angles, even though this arrangement gives a low filling of space, 0.34 of the available space, compared with 0.74 for a close-packed structure. The tetrahedral bond allows only four nearest neighbors, whereas a close-packed structure has 12. We should not overemphasize the similarity of the bonding of carbon and silicon. Carbon gives biology, but silicon gives geology and semiconductor technology.

The binding of molecular hydrogen is a simple example of a covalent bond. The strongest binding (Fig. 12) occurs when the spins of the two electrons are antiparallel. The binding depends on the relative spin orientation not because there are strong magnetic dipole forces between the spins, but because the Pauli principle modifies the distribution of charge according to the spin orientation. This spin-dependent coulomb energy is called the **exchange interaction**.

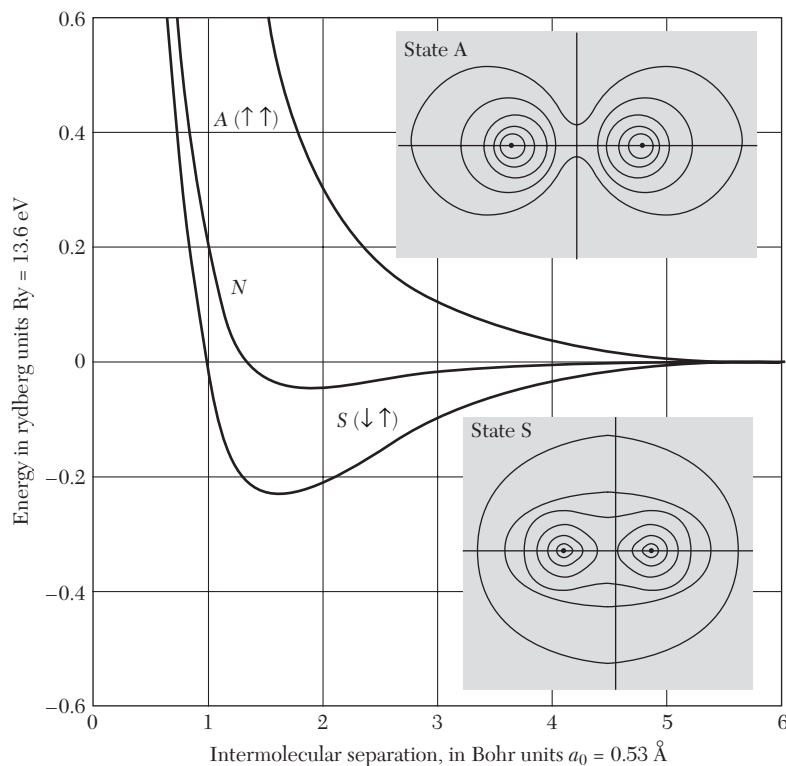


Figure 12 Energy of molecular hydrogen (H_2) referred to separated neutral atoms. A negative energy corresponds to binding. The curve N refers to a classical calculation with free atom charge densities; A is the result for parallel electron spins, taking the Pauli exclusion principle into account, and S (the stable state) for antiparallel spins. The density of charge is represented by contour lines for the states A and S .

The Pauli principle gives a strong repulsive interaction between atoms with filled shells. If the shells are not filled, electron overlap can be accommodated without excitation of electrons to high energy states and the bond will be shorter. Compare the bond length ($2 \text{ \AA}$) of Cl_2 with the interatomic distance ($3.76 \text{ \AA}$) of Ar in solid Ar ; also compare the cohesive energies given in Table 1. The difference between Cl_2 and Ar_2 is that the Cl atom has five electrons in the $3p$ shell and the Ar atom has six, filling the shell, so that the repulsive interaction is stronger in Ar than in Cl .

The elements C , Si , and Ge lack four electrons with respect to filled shells, and thus these elements (for example) can have an attractive interaction associated with charge overlap. The electron configuration of carbon is $1s^2 2s^2 2p^2$. To form a tetrahedral system of covalent bonds the carbon atom must first be promoted to the electronic configuration $1s^2 2s^2 2p^3$. This promotion from the ground state requires 4 eV , an amount more than regained when the bonds are formed.